

Supplementary material 3

The Big Picture: a Comparative Analysis of the Social Determinants of Health Across Europe

```
# Libraries loading
library(haven)

## Warning: package 'haven' was built under R version 4.4.1
library(dplyr)

## Warning: package 'dplyr' was built under R version 4.4.1
##
## Adjuntando el paquete: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.4.3
library(tidyr)

## Warning: package 'tidyr' was built under R version 4.4.1
library(lme4)

## Warning: package 'lme4' was built under R version 4.4.1
## Cargando paquete requerido: Matrix
##
## Adjuntando el paquete: 'Matrix'
## The following objects are masked from 'package:tidyr':
##
##   expand, pack, unpack
library(readxl)

## Warning: package 'readxl' was built under R version 4.4.1
library(labelled)

## Warning: package 'labelled' was built under R version 4.4.3
```

Data arrangements

Loading of the original dataset.

```
# Data downloaded from: https://doi.org/10.21338/ess11e04\_1
ESS11FullRaw <- read.csv(file.path("datasets_raw", "ESS11e04_1.csv"))
```

Recoding of some of the variables of analysis

```
# NAs for some variables
ESS11FullRaw$cgtsmok[ESS11FullRaw$cgtsmok == 7 | ESS11FullRaw$cgtsmok == 8 | ESS11FullRaw$cgtsmok == 9]
ESS11FullRaw$etfruit[ESS11FullRaw$etfruit == 77 | ESS11FullRaw$etfruit == 88 | ESS11FullRaw$etfruit == 99]
ESS11FullRaw$eatveg[ESS11FullRaw$eatveg == 77 | ESS11FullRaw$eatveg == 88 | ESS11FullRaw$eatveg == 99]
ESS11FullRaw$mbtru[ESS11FullRaw$mbtru == 7 | ESS11FullRaw$mbtru == 8 | ESS11FullRaw$mbtru == 9] <- NA
ESS11FullRaw$hincfel[ESS11FullRaw$hincfel == 7 | ESS11FullRaw$hincfel == 8 | ESS11FullRaw$hincfel == 9]
ESS11FullRaw$alcfreq[ESS11FullRaw$alcfreq == 77 | ESS11FullRaw$alcfreq == 88 | ESS11FullRaw$alcfreq == 99]
ESS11FullRaw$medtrun[ESS11FullRaw$medtrun == 7 | ESS11FullRaw$medtrun == 8 | ESS11FullRaw$medtrun == 9]
ESS11FullRaw$volunfp[ESS11FullRaw$volunfp == 7 | ESS11FullRaw$volunfp == 8 | ESS11FullRaw$volunfp == 9]
# Sex ----

ESS11Rec <- ESS11FullRaw %>%
  mutate(
    gender = case_when(
      gndr == 1 ~ 0, # Male becomes 0
      gndr == 2 ~ 1 # Female becomes 1
    )
  ) %>%
  mutate(
    gender = set_value_labels(
      gender,
      labels = c("Male" = 0, "Female" = 1)
    )
  )

# Age ----

ESS11Rec <- ESS11Rec %>%
  mutate(
    age = 2024 - yrbrn
  )

# RegionWelfare ----

ESS11Rec <- ESS11Rec %>%
  mutate(regionWelfare = case_when(
    cntry %in% c("FI", "IS", "NO", "SE") ~ 1, # Scandinavian
    cntry %in% c("GB", "IE") ~ 2, # Anglo-Saxon
    cntry %in% c("AT", "BE", "CH", "DE", "FR", "IL", "NL") ~ 3, # Bismarkian
    cntry %in% c("BG", "EE", "HU", "HR", "LT", "LV", "ME", "PL", "RS", "SI", "SK", "UA") ~ 4, # Eastern
    cntry %in% c("ES", "GR", "IT", "PT", "CY") ~ 5, # Southern
    TRUE ~ 99
  )) %>%
```

```

mutate(regionWelfare = set_value_labels(
  regionWelfare,
  labels = c("Scandinavian" = 1, "Anglo-Saxon" = 2, "Mismarkian" = 3, "Eastern European " = 4, "Souther
))

# Education ----

## eiscedV
ESS11Rec <- ESS11Rec %>%
  mutate(
    eiscedV = case_when(
      eisced == 0 ~ eisced,
      eisced == 1 ~ eisced,
      eisced == 2 ~ eisced,
      eisced == 3 ~ eisced,
      eisced == 4 ~ 3,
      eisced == 5 ~ 4,
      eisced == 6 ~ 5,
      eisced == 7 ~ 5,
      eisced == 55 ~ NA_real_
    )
  ) %>%
  mutate(
    eiscedV = set_value_labels(
      eiscedV,
      labels = c("Not possible to harmonise into ES-ISCED" = 0, "ES-ISCED I , less than lower secondary
        "ES-ISCED II, lower secondary" = 2, "ES-ISCED III" = 3, "ISCED IV and ISCED v" = 4, "I
        "Other" = 55, "refusal" = 77, "Don't know" = 88, "No answer" = 99 )
    )
  )

# eisced2
ESS11Rec <- ESS11Rec %>%
  mutate( eisced2 = if_else(eiscedV < 3, 1, 0)) #eisced2 1 = ES-ISCED IIIb, lower tier upper seconda

# Health behaviours ----

## Smoking ----
ESS11Rec <- ESS11Rec %>%
  mutate(
    cgtsmoker = case_when(
      cgtsmok == 1 ~ 5,
      cgtsmok == 2 ~ 4,
      cgtsmok == 3 ~ 3,
      cgtsmok == 4 ~ 2,
      cgtsmok == 5 ~ 1,
      cgtsmok == 6 ~ 0,
      cgtsmok == 7 ~ 7,
      cgtsmok == 8 ~ 8
    ) #cgtsmoke = 5 means i smoke daily
  ) %>%
  mutate(
    cgtsmoker = set_value_labels(

```



```

ESS11Rec <- ESS11Rec %>%
  mutate(ctrllife = case_when(
    ctrllife == 0 ~ 10,
    ctrllife == 1 ~ 9,
    ctrllife == 2 ~ 8,
    ctrllife == 3 ~ 7,
    ctrllife == 4 ~ 6,
    ctrllife == 5 ~ 5,
    ctrllife == 6 ~ 4,
    ctrllife == 7 ~ 3,
    ctrllife == 8 ~ 2,
    ctrllife == 9 ~ 1,
    ctrllife == 10 ~ 0,
  )) # 0 is complete cntrl, 10 is no ctrl

## Life satisfaction ----
ESS11Rec <- ESS11Rec %>%
  mutate(stflifer = case_when(
    stflife == 0 ~ 10,
    stflife == 1 ~ 9,
    stflife == 2 ~ 8,
    stflife == 3 ~ 7,
    stflife == 4 ~ 6,
    stflife == 5 ~ 5,
    stflife == 6 ~ 4,
    stflife == 7 ~ 3,
    stflife == 8 ~ 2,
    stflife == 9 ~ 1,
    stflife == 10 ~ 0,
  )) %>% # 0 is Extremely satisfied, 10 is Extremely dissatisfied
  mutate(stflifer01 = stflifer/10)

## Childhood conflicts ----
ESS11Rec <- ESS11Rec %>%
  mutate( #conflict
    cnflpplh = case_when(
      cnfpplh == 1 ~ 4,
      cnfpplh == 2 ~ 3,
      cnfpplh == 3 ~ 2,
      cnfpplh == 4 ~ 1,
      cnfpplh == 5 ~ 0,
    ) # cnflpplh = 4 means always
  ) %>%
  mutate(
    cnflpplh = set_value_labels(
      cnflpplh, labels = c("Never" = 0, "Hardly ever" = 1, "Sometimes" = 2, "Often" = 3, "Always" = 4,
        "Refusal" = 7, "Don't Know" = 8, "No answer" = 9)
    )
  )

# Employment conditions ----

```

```

## Unemployment ----
ESS11Rec <- ESS11Rec %>%
  mutate(unemplD = if_else((uempli == 1)|(uempla == 1) , 1, 0)) # unemplD 1 means unemployed, either lo

## Ergonomic ----
ESS11Rec <- ESS11Rec %>%
  mutate(ergonomic = rowSums(across(
    c(jbexpmc, jbexpml, jbexpti, jbexpvi),
    ~ .x == 1),
    na.rm = TRUE))

## Material ----
ESS11Rec <- ESS11Rec %>%
  mutate(material = rowSums(across(c(jbexebs, jbexecp, jbexera, jbexevc, jbexevh, jbexevl),
    ~ .x == 1),
    na.rm = TRUE))

## Job control ----
ESS11Rec <- ESS11Rec %>%
  mutate(wkorg = case_when(
    wkdcorga == 0 ~ 10,
    wkdcorga == 1 ~ 9,
    wkdcorga == 2 ~ 8,
    wkdcorga == 3 ~ 7,
    wkdcorga == 4 ~ 6,
    wkdcorga == 5 ~ 5,
    wkdcorga == 6 ~ 4,
    wkdcorga == 7 ~ 3,
    wkdcorga == 8 ~ 2,
    wkdcorga == 9 ~ 1,
    wkdcorga == 10 ~ 0,
  ))

## Job organization decisions ----
ESS11Rec <- ESS11Rec %>%
  mutate(iorgactr = case_when(
    iorgact == 0 ~ 10,
    iorgact == 1 ~ 9,
    iorgact == 2 ~ 8,
    iorgact == 3 ~ 7,
    iorgact == 4 ~ 6,
    iorgact == 5 ~ 5,
    iorgact == 6 ~ 4,
    iorgact == 7 ~ 3,
    iorgact == 8 ~ 2,
    iorgact == 9 ~ 1,
    iorgact == 10 ~ 0,
  ))

# member of a trade union or similar ----

```

```

ESS11Rec <- ESS11Rec %>%
  mutate (mbtruD = if_else(mbtru == 3 | mbtru == 2, 1, 0)) # no, or not currently

# Material conditions ----

## Financial difficulties growing up ----

ESS11Rec <- ESS11Rec %>%
  mutate(
    fndf = case_when(
      fnsdfml == 1 ~ 4,
      fnsdfml == 2 ~ 3,
      fnsdfml == 3 ~ 2,
      fnsdfml == 4 ~ 1,
      fnsdfml == 5 ~ 0,
    ) # fndf 4 means Always inancial difficulties growing up
  ) %>%
  mutate(
    fndf = set_value_labels(
      fndf, labels = c("Never" = 0, "Hardly ever" = 1, "Sometimes" = 2, "Often" = 3, "Always" = 4, "Refus
    )
  )

## Housing problems ----
ESS11Rec <- ESS11Rec %>%
  mutate(housing = rowSums(across(
    c(paccmoro, paccdwlr, pacclift, paccnbsh, paccocrw, paccxhoc, paccnois, paccinro),
    ~ .x == 1),
    na.rm = TRUE))

# Social Capital ----

## Meeting people ----
ESS11Rec <- ESS11Rec %>%
  mutate(socialmeet = case_when(
    sclmeet == 1 ~ 6,
    sclmeet == 2 ~ 5,
    sclmeet == 3 ~ 4,
    sclmeet == 4 ~ 3,
    sclmeet == 5 ~ 2,
    sclmeet == 6 ~ 1,
    sclmeet == 7 ~ 0,
  )) # risk! socialmeet 0 = means everyday meetings, sclmeet 6 == never

## people with whom to talk ----
ESS11Rec <- ESS11Rec %>%
  mutate(inprdsc = case_when(
    inprdsc == 0 ~ 6,
    inprdsc == 1 ~ 5,
    inprdsc == 2 ~ 4,

```

```

inprdsc == 3 ~ 3,
inprdsc == 4 ~ 2,
inprdsc == 5 ~ 1,
inprdsc == 6 ~ 0,
)) # risk! inperdisc 0 >= 10 ppl to talk, inperdisc 6 = none

## volunteered ----
ESS11Rec <- ESS11Rec %>%
  mutate (volunfpr = if_else(volunfp == 2, 1, 0))

## People can be trusted ----
ESS11Rec <- ESS11Rec %>%
  mutate(ppltrst = case_when(
    ppltrst == 0 ~ 10,
    ppltrst == 1 ~ 9,
    ppltrst == 2 ~ 8,
    ppltrst == 3 ~ 7,
    ppltrst == 4 ~ 6,
    ppltrst == 5 ~ 5,
    ppltrst == 6 ~ 4,
    ppltrst == 7 ~ 3,
    ppltrst == 8 ~ 2,
    ppltrst == 9 ~ 1,
    ppltrst == 10 ~ 0,
  )) # risk! ppltrst 0 is      Most people can be trusted, 10 is You can't be too careful

## Politicians can be trusted ----
ESS11Rec <- ESS11Rec %>%
  mutate(trstplt = case_when(
    trstplt == 0 ~ 10,
    trstplt == 1 ~ 9,
    trstplt == 2 ~ 8,
    trstplt == 3 ~ 7,
    trstplt == 4 ~ 6,
    trstplt == 5 ~ 5,
    trstplt == 6 ~ 4,
    trstplt == 7 ~ 3,
    trstplt == 8 ~ 2,
    trstplt == 9 ~ 1,
    trstplt == 10 ~ 0,
  ))

# Healthcare ----

## Unpaid care ----
ESS11Rec <- ESS11Rec %>%
  mutate(hlpfmhrr = case_when( ## helps, and does it more than 10hrs/week
    hlpfmhr == 1 ~ hlpfmhr,
    hlpfmhr == 2 ~ hlpfmhr,

```



```

    hlpfmhr == 3 ~ hlpfmhr,
    hlpfmhr == 4 ~ hlpfmhr,
    hlpfmhr == 5 ~ hlpfmhr,
    hlpfmhr == 6 ~ hlpfmhr,
    hlpfmhr == 55 ~ 0,
  ))

ESS11Rec <- ESS11Rec %>%
  mutate(
    hlpfmhrrcomb = case_when(
      hlpfmly == 2 ~ 0,
      hlpfmhrr == 0 ~ hlpfmhrr,
      hlpfmhrr > 0 ~ hlpfmhrr )
  )

## HC unmet need ----
ESS11Rec <- ESS11Rec %>%
  mutate(
    medtrun2 = case_when(
      medtrun == 1 ~ 1,
      medtrun == 2 ~ 0,
      medtrun == 7 ~ 7,
      medtrun == 8 ~ 8,
    )
  ) %>%
  mutate(
    medtrun2 = set_value_labels(
      medtrun2, labels = c("unmet need" = 1, "No unmet need" = 0, "Refusal" = 7, "Don't Know" = 8)
    )
  )

## General Practitioner ----
ESS11Rec <- ESS11Rec %>%
  mutate(dshltgpr = if_else(dshltgpr == 0, 1, 0)) #not discussed, dshltgpr = 1, discussed = 0

## Medical specialist ----
ESS11Rec <- ESS11Rec %>%
  mutate(dshltmsr = if_else(dshltms == 0, 1, 0))

# Identification number per row ----
ESS11Rec <- ESS11Rec %>%
  mutate(id_row = row_number())

# Deletion of Israel
ESS11RecEu <- ESS11Rec %>%
  filter(cntry != "IL")

# Produce a vector of all countries
country_list <- unique(ESS11RecEu$cntry)

```

```

# Prepare region and regionWelfare datasets to join later with the Block_scores datasets
country_regionWelfare <- ESS11RecEu %>%
  dplyr::select(cntry, regionWelfare) %>%
  distinct()

WR_list <- country_regionWelfare %>%
  distinct(regionWelfare) %>%
  filter(regionWelfare != 99) %>%
  pull(regionWelfare) %>%
  sort()
WR_labels <- c("1" = "Scandinavian", "2" = "Anglo-Saxon", "3" = "Bismarckian",
               "4" = "Eastern European", "5" = "Southern")

```

Statistical analysis

Additional libraries for statistical analysis

```

library(survey)

## Warning: package 'survey' was built under R version 4.4.1
## Cargando paquete requerido: grid
## Cargando paquete requerido: survival
## Warning: package 'survival' was built under R version 4.4.1
##
## Adjuntando el paquete: 'survey'
## The following object is masked from 'package:graphics':
##
##      dotchart

library(haven)
library(dplyr)
library(ggplot2)
library(tidyr)
library(lme4)
library(readxl)
library(labelled)
library(openxlsx)

## Warning: package 'openxlsx' was built under R version 4.4.3

library(ltm)

## Warning: package 'ltm' was built under R version 4.4.3
## Cargando paquete requerido: MASS
## Warning: package 'MASS' was built under R version 4.4.3
##
## Adjuntando el paquete: 'MASS'
## The following object is masked from 'package:dplyr':
##
##      select

```

```

## Cargando paquete requerido: msm
## Warning: package 'msm' was built under R version 4.4.3
## Cargando paquete requerido: polycor
## Warning: package 'polycor' was built under R version 4.4.3
library(psych)

## Warning: package 'psych' was built under R version 4.4.1
##
## Adjuntando el paquete: 'psych'
## The following object is masked from 'package:ltm':
##
##     factor.scores
## The following object is masked from 'package:polycor':
##
##     polyserial
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
library(ggplot2)
library(forcats)

## Warning: package 'forcats' was built under R version 4.4.1
library(fmsb)

## Warning: package 'fmsb' was built under R version 4.4.3
library(janitor)

## Warning: package 'janitor' was built under R version 4.4.3
##
## Adjuntando el paquete: 'janitor'
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
library(effectsize)

## Warning: package 'effectsize' was built under R version 4.4.3
##
## Adjuntando el paquete: 'effectsize'
## The following objects are masked from 'package:fmsb':
##
##     oddsratio, riskratio
## The following object is masked from 'package:psych':
##
##     phi
library(car)

```

```
## Warning: package 'car' was built under R version 4.4.1
## Cargando paquete requerido: carData
## Warning: package 'carData' was built under R version 4.4.1
##
## Adjuntando el paquete: 'car'
## The following object is masked from 'package:psych':
##
##     logit
## The following object is masked from 'package:dplyr':
##
##     recode
library(dendextend)

## Warning: package 'dendextend' was built under R version 4.4.3
##
## -----
## Welcome to dendextend version 1.19.1
## Type citation('dendextend') for how to cite the package.
##
## Type browseVignettes(package = 'dendextend') for the package vignette.
## The github page is: https://github.com/talgalili/dendextend/
##
## Suggestions and bug-reports can be submitted at: https://github.com/talgalili/dendextend/issues
## You may ask questions at stackoverflow, use the r and dendextend tags:
##   https://stackoverflow.com/questions/tagged/dendextend
##
## To suppress this message use: suppressPackageStartupMessages(library(dendextend))
## -----
##
## Adjuntando el paquete: 'dendextend'
## The following object is masked from 'package:stats':
##
##     cutree
```

PCA: Composite risk score

Preparations

```
# Create a DataSet
pca_variables <- c("id_row", "cntry", "regionWelfare", "psu", "stratum", "gender", "pspwght", "anweight")

# Check NAs
colSums(is.na(ESS11RecEu[,pca_variables]))
```

	id_row	cntry	regionWelfare	psu	stratum
##	0	0	0	0	0
	gender	pspwght	anweight	eiscsd2	material
##	0	0	0	322	0
	ergonomic	wkorg	iorgactr	unemplD	mbtruD
##	0	5307	5450	0	296

```
##      housing      fndf      hincfel      medtrun2  hlpfmhrrcomb
##      0          759      653          393          617
##      dshltgpr      dshltmsr      etfruitveg      dosportr      cgtsmoker
##      0          0          240          769          112
##      alcoholfreq      cntrlife      stflifer      conflpplh      socialmeet
##      345          282          360          663          132
##      inperdisc      ppltrust      trustplt      volunfpr
##      309          141          751          166

# Make dataset
pca_df <- ESS11RecEu %>%
  dplyr::select(all_of(pca_variables))

# Include sex as factor
pca_df <- pca_df %>%
  mutate(sex = factor(gender, levels = c(0,1), labels = c("Men", "Women")))
```

Education

```
#Step 0: Make a df only for the subset of variables (to drop NAs)
index_df_edu <- pca_df %>%
  drop_na(c("id_row", "eiscsed2"))

#Step 1: Calculate Z-Scores = standardized comparative indicator
index_df_edu_scores <- index_df_edu %>%
  mutate(
    index_edu = as.numeric(scale(eiscsed2))
  )

#Step 2: Define survey design
index_design_edu <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~pspwght,
  data = index_df_edu_scores,
  nest = TRUE
)
options(survey.lonely.psu = "adjust")

# Step 3: Estimate means with confidence intervals

#By sex
edu_index_bysex <- svyby(
  ~index_edu,
  ~cuntry + sex,
  index_design_edu,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
)

#pooled sex
edu_index_pooledsex <- svyby(
```

```

~index_edu,
~cntry,
index_design_edu,
svymean,
na.rm = TRUE,
vartype = c("se", "ci")
) %>%
  as.data.frame() %>%
  mutate(sex = "All")

# Europe pooled (mean of countries)
edu_index_pooledCountry <- mean(edu_index_pooledsex$index_edu, na.rm = TRUE)

# WR pooled (mean of countries)
edu_index_pooledWR <- edu_index_pooledsex %>%
  left_join(country_regionWelfare, by = "cntry") %>%
  group_by(regionWelfare) %>%
  summarise(
    wr_mean_edu = mean(index_edu, na.rm = TRUE),
    .groups = "drop"
  )

# Step 4: Bind analyses, and add region info back in
edu_index_score <- bind_rows(
  edu_index_pooledsex %>% dplyr::select(cntry, index_edu, se, ci_l, ci_u, sex),
  edu_index_bysex %>% dplyr::select(cntry, index_edu, se, ci_l, ci_u, sex)
)

edu_index_score <- edu_index_score %>%
  left_join(country_regionWelfare, by = "cntry")

```

Employment Conditions

```

#Step 0: Make a df only for the subset of variables (to drop NAs)
pca_df_work <- pca_df %>%
  drop_na(c("unemplD", "mbtruD", "material", "ergonomic", "wkorg", "iorgactr"))

# Calculate a manual reference Risk score to know the direction of risk (and correct the PCA sign if ne
risk_ref_work <- pca_df_work %>%
  mutate(across(
    c("unemplD", "mbtruD", "material", "ergonomic", "wkorg", "iorgactr"),
    ~ (.-min(., na.rm=TRUE)) / (max(., na.rm=TRUE) - min(., na.rm=TRUE))
  )) %>%
  transmute(risk_ref_psych = rowMeans(across(c("unemplD", "mbtruD", "material", "ergonomic", "wkorg", "iorgactr"),
    pull()

# Step 1: Run PCA
pca_work <- prcomp( pca_df_work[, c("material", "ergonomic", "unemplD", "wkorg", "iorgactr", "mbtruD")],
  scale. = TRUE
)

```

```

# Check if sign is aligned with risk reference
cor_work <- cor(pca_work$x[,1], risk_ref_work, use = "complete.obs")

if(cor_work < 0){
  pca_work$x[,1] <- -pca_work$x[,1]
  pca_work$rotation[,1] <- -pca_work$rotation[,1]
}

print(pca_work)

## Standard deviations (1, .., p=6):
## [1] 1.3159131 1.2910547 1.0021383 0.9631858 0.5813582 0.5758167
##
## Rotation (n x k) = (6 x 6):
##
##          PC1          PC2          PC3          PC4          PC5
## material -0.4321660  0.54196704 -0.03855641 -0.133848555 -0.69143519
## ergonomic -0.4152517  0.55576839 -0.04670759 -0.129739996  0.69327429
## unemplD   0.1006252  0.17645302 -0.73363261  0.648370008 -0.01037709
## wkorg     0.5464943  0.41999951  0.14542091 -0.026468919  0.13412504
## iorgactr  0.5308012  0.43250693  0.18898354  0.005441824 -0.15184886
## mbtruD    0.2242035 -0.05287463 -0.63344533 -0.737657246 -0.01152632
##
##          PC6
## material  0.148390209
## ergonomic -0.138000734
## unemplD   -0.005545296
## wkorg     0.696496215
## iorgactr  -0.687302984
## mbtruD    -0.037590206

# Merge the result of the PCA to the initial pca_df
pca_df_work_scores <- pca_df_work %>%
  mutate(pc1_work = pca_work$x[,1])

# Step 2: Define survey design to be able to do survey-weighted calculations later on
pca_design_work <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~pspwght,
  data = pca_df_work_scores,
  nest = TRUE
)
options(survey.lonely.psu = "adjust")

# Step 3: Estimate means with confidence intervals

#By sex
work_pca_bysex <- svyby(
  ~pc1_work,
  ~cuntry + sex,
  pca_design_work,
  svymean,
  na.rm = TRUE,

```

```

  vartype = c("se", "ci")
)

#pooled sex
work_pca_pooledsex <- svyby(
  ~pc1_work,
  ~cntry,
  pca_design_work,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
) %>%
  as.data.frame() %>%
  mutate(sex = "All")

# Europe pooled (mean of countries)
work_pca_pooledCountry <- mean(work_pca_pooledsex$pc1_work, na.rm = TRUE)

# WR pooled (mean of countries)
work_pca_pooledWR <- work_pca_pooledsex %>%
  left_join(country_regionWelfare, by = "cntry") %>%
  group_by(regionWelfare) %>%
  summarise(
    wr_mean_work = mean(pc1_work, na.rm = TRUE),
    .groups = "drop"
  )

# Step 4: Bind analyses, and add region info back in (once per country)
work_pca_score <- bind_rows(
  work_pca_pooledsex %>% dplyr::select(cntry, pc1_work, se, ci_l, ci_u, sex),
  work_pca_bysex %>% dplyr::select(cntry, pc1_work, se, ci_l, ci_u, sex)
)

work_pca_score <- work_pca_score %>%
  left_join(country_regionWelfare, by = "cntry")

```

Material circumstances

```

#Step 0: Make a df only for the subset of variables (to drop NAs)
pca_df_mat <- pca_df %>%
  drop_na(c("id_row", "housing", "fndf", "hincfel"))

# Calculate a manual reference Risk score to know the direction of risk (to correct the PCA sign if needed)
risk_ref_mat <- pca_df_mat %>%
  mutate(across(
    c("housing", "fndf", "hincfel"),
    ~ (.-min(., na.rm=TRUE)) / (max(., na.rm=TRUE) - min(., na.rm=TRUE))
  )) %>%
  transmute(risk_ref_psych = rowMeans(across(c("housing", "fndf", "hincfel")), na.rm = TRUE)) %>%
  pull()

```



```

# Step 1: Run PCA

pca_mat <- prcomp(pca_df_mat[, c("housing", "fndf", "hincfel")],
                  scale. = TRUE
)

print(pca_mat)

## Standard deviations (1, ..., p=3):
## [1] 1.1927145 0.9479852 0.8238666
##
## Rotation (n x k) = (3 x 3):
##           PC1      PC2      PC3
## housing 0.4439276 0.8911773 -0.09344143
## fndf     0.6234758 -0.3820966 -0.68211446
## hincfel 0.6435886 -0.2445510 0.72525069

# Check if sign is aligned with risk reference

cor_mat <- cor(pca_mat$x[,1], risk_ref_mat, use = "complete.obs")

if(cor_mat < 0){
  pca_mat$x[,1] <- -pca_mat$x[,1]
  pca_mat$rotation[,1] <- -pca_mat$rotation[,1]
}

print(pca_mat)

## Standard deviations (1, ..., p=3):
## [1] 1.1927145 0.9479852 0.8238666
##
## Rotation (n x k) = (3 x 3):
##           PC1      PC2      PC3
## housing 0.4439276 0.8911773 -0.09344143
## fndf     0.6234758 -0.3820966 -0.68211446
## hincfel 0.6435886 -0.2445510 0.72525069

# Merge the result of the PCA to the initial pca_df
pca_df_mat_scores <- pca_df_mat %>%
  mutate(pc1_mat = pca_mat$x[,1])

# Step 2: Define survey design
pca_design_mat <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~pspwght,
  data = pca_df_mat_scores,
  nest = TRUE
)

options(survey.lonely.psu = "adjust")

# Step 3: Estimate means with confidence intervals

```

```

#By sex
mat_pca_bysex <- svyby(
  ~pc1_mat,
  ~cntry + sex,
  pca_design_mat,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
)

#pooled sex
mat_pca_pooledsex <- svyby(
  ~pc1_mat,
  ~cntry,
  pca_design_mat,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
) %>%
  as.data.frame() %>%
  mutate(sex = "All")

# Europe pooled (mean of countries)

mat_pca_pooledCountry <- mean(mat_pca_pooledsex$pc1_mat, na.rm = TRUE)

# WR pooled (mean of countries)

mat_pca_pooledWR <- mat_pca_pooledsex %>%
  left_join(country_regionWelfare, by = "cntry") %>%
  group_by(regionWelfare) %>%
  summarise(
    wr_mean_mat = mean(pc1_mat, na.rm = TRUE),
    .groups = "drop"
  )

# Step 4: Bind analyses, and add region info back in (once per country)

mat_pca_score <- bind_rows(
  mat_pca_pooledsex %>% dplyr::select(cntry, pc1_mat, se, ci_l, ci_u, sex),
  mat_pca_bysex %>% dplyr::select(cntry, pc1_mat, se, ci_l, ci_u, sex)
)

mat_pca_score <- mat_pca_score %>%
  left_join(country_regionWelfare, by = "cntry")

```

HC access

```

# Step 0: Make a df only for the subset of variables (to drop NAs)
pca_df_hc <- pca_df %>%
  drop_na(c("medtrun2", "hlpfmhrrcomb", "dshltmsr", "dshltgpr"))

```

```

# Calculate a manual reference Risk score to know the direction of risk (and correct the PCA sign if ne
risk_ref_hc <- pca_df_hc %>%
  mutate(across(c("medtrun2", "dshltmsr"),
    ~ (-min(., na.rm=TRUE)) / (max(., na.rm=TRUE) - min(., na.rm=TRUE))
  )) %>%
  transmute(risk_ref_hc = rowMeans(across(c("medtrun2")), na.rm = TRUE)) %>%
  pull()

# Step 1: Run PCA
pca_hc <- prcomp(pca_df_hc[, c( "medtrun2", "hlpfmhrrcomb", "dshltmsr", "dshltgpr")],
  scale. = TRUE)

print(pca_hc)

## Standard deviations (1, .., p=4):
## [1] 1.1521685 0.9923187 0.9379815 0.8988893
##
## Rotation (n x k) = (4 x 4):
##           PC1          PC2          PC3          PC4
## medtrun2    0.5245951 -0.07253681 -0.8184818  0.2227687
## hlpfmhrrcomb 0.2494275 -0.93836471  0.2387746 -0.0156295
## dshltmsr    -0.5927871 -0.20942917 -0.1539094  0.7622695
## dshltgpr    -0.5578435 -0.26523481 -0.4993857 -0.6075154

#Check if sign is aligned with risk reference
cor_hc <- cor(pca_hc$x[,1], risk_ref_hc, use = "complete.obs")

if(cor_hc < 0){
  pca_hc$x[,1] <- -pca_hc$x[,1]
  pca_hc$rotation[,1] <- -pca_hc$rotation[,1]
}
print(pca_hc)

## Standard deviations (1, .., p=4):
## [1] 1.1521685 0.9923187 0.9379815 0.8988893
##
## Rotation (n x k) = (4 x 4):
##           PC1          PC2          PC3          PC4
## medtrun2    0.5245951 -0.07253681 -0.8184818  0.2227687
## hlpfmhrrcomb 0.2494275 -0.93836471  0.2387746 -0.0156295
## dshltmsr    -0.5927871 -0.20942917 -0.1539094  0.7622695
## dshltgpr    -0.5578435 -0.26523481 -0.4993857 -0.6075154

# Merge the result of the PCA to the initial pca_df
pca_df_hc_scores <- pca_df_hc %>%
  mutate(pc1_hc = pca_hc$x[,1])

# Step 2: Define survey design
pca_design_hc <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~pspwght,
  data = pca_df_hc_scores,
  nest = TRUE

```

```

)

# Step 3: Estimate means with confidence intervals

#By sex
hc_pca_bysex <- svyby(
  ~pc1_hc,
  ~cntry + sex,
  pca_design_hc,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
)

#pooled sex
hc_pca_pooledsex <- svyby(
  ~pc1_hc,
  ~cntry,
  pca_design_hc,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
) %>%
  as.data.frame() %>%
  mutate(sex = "All")

# Europe pooled (mean of countries)

hc_pca_pooledCountry <- mean(hc_pca_pooledsex$pc1_hc, na.rm = TRUE)

# WR pooled (mean of countries)

hc_pca_pooledWR <- hc_pca_pooledsex %>%
  left_join(country_regionWelfare, by = "cntry") %>%
  group_by(regionWelfare) %>%
  summarise(
    wr_mean_hc = mean(pc1_hc, na.rm = TRUE),
    .groups = "drop"
  )

# Step 4: Bind analyses, and add region info back in (once per country)
hc_pca_score <- bind_rows(
  hc_pca_pooledsex %>% dplyr::select(cntry, pc1_hc, se, ci_l, ci_u, sex),
  hc_pca_bysex %>% dplyr::select(cntry, pc1_hc, se, ci_l, ci_u, sex)
)

hc_pca_score <- hc_pca_score %>%
  left_join(country_regionWelfare, by = "cntry")

```

Behavioural

```
# Step 0: Make a df only for the subset of variables (to drop NAs)
pca_df_behav <- pca_df %>%
  drop_na(c("id_row", "etfruitveg", "dosportr", "cgtsmoker", "alcoholfreq")
) )

# Calculate a manual reference Risk score to know the direction of risk (and correct the PCA sign if ne

risk_ref_behav <- pca_df_behav %>%
  mutate(across(
    c("etfruitveg", "dosportr", "cgtsmoker", "alcoholfreq"),
    ~ (.-min(., na.rm=TRUE)) / (max(., na.rm=TRUE) - min(., na.rm=TRUE))
  )) %>%
  transmute(risk_ref_behav = rowMeans(across(c("etfruitveg", "dosportr", "cgtsmoker", "alcoholfreq")), na.rm=TRUE),
    pull())

# Step 1: Run PCA

pca_behav <- prcomp(pca_df_behav[, c("etfruitveg", "dosportr", "cgtsmoker", "alcoholfreq")],
  scale. = TRUE)

print(pca_behav)

## Standard deviations (1, ..., p=4):
## [1] 1.1400331 1.0648238 0.9067568 0.8627091
##
## Rotation (n x k) = (4 x 4):
##          PC1      PC2      PC3      PC4
## etfruitveg 0.5582083 0.3543173 0.65815161 0.3601377
## dosportr   0.3159248 0.6489084 -0.68200064 0.1182562
## cgtsmoker  0.6519775 -0.2239503 -0.03651693 -0.7234903
## alcoholfreq 0.4043765 -0.6349987 -0.31682503 0.5769559

#Check if sign is aligned with risk reference
cor_behav <- cor(pca_behav$x[,1], risk_ref_behav, use = "complete.obs")

if(cor_behav < 0){
  pca_behav$x[,1] <- -pca_behav$x[,1]
  pca_behav$rotation[,1] <- -pca_behav$rotation[,1]
}
print(pca_behav)

## Standard deviations (1, ..., p=4):
## [1] 1.1400331 1.0648238 0.9067568 0.8627091
##
## Rotation (n x k) = (4 x 4):
##          PC1      PC2      PC3      PC4
## etfruitveg 0.5582083 0.3543173 0.65815161 0.3601377
## dosportr   0.3159248 0.6489084 -0.68200064 0.1182562
## cgtsmoker  0.6519775 -0.2239503 -0.03651693 -0.7234903
## alcoholfreq 0.4043765 -0.6349987 -0.31682503 0.5769559
```

```

# Merge the result of the PCA to the initial pca_df

pca_df_behav_scores <- pca_df_behav %>%
  mutate(pc1_behav = pca_behav$x[,1])

# Step 2: Define survey design
pca_design_behav <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~pspwght,
  data = pca_df_behav_scores,
  nest = TRUE
)

# Step 3: Estimate means with confidence intervals

#By sex
behav_pca_bysex <- svyby(
  ~pc1_behav,
  ~centry + sex,
  pca_design_behav,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
)

#pooled sex
behav_pca_pooledsex <- svyby(
  ~pc1_behav,
  ~centry,
  pca_design_behav,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
) %>%
  as.data.frame() %>%
  mutate(sex = "All")

# Europe pooled (mean of countries)

behav_pca_pooledCountry <- mean(behav_pca_pooledsex$pc1_behav, na.rm = TRUE)

# WR pooled (mean of countries)

behav_pca_pooledWR <- behav_pca_pooledsex %>%
  left_join(country_regionWelfare, by = "centry") %>%
  group_by(regionWelfare) %>%
  summarise(
    wr_mean_behav = mean(pc1_behav, na.rm = TRUE),
    .groups = "drop"
  )

```

```

)

# Step 4: Bind analyses

behav_pca_score <- bind_rows(
  behav_pca_pooledsex %>% dplyr::select(cntry, pc1_behav, se, ci_l, ci_u, sex),
  behav_pca_bysex %>% dplyr::select(cntry, pc1_behav, se, ci_l, ci_u, sex)
)

behav_pca_score <- behav_pca_score %>%
  left_join(country_regionWelfare, by = "cntry")

```

Psychosocial

```

# Step 0: Make a df only for the subset of variables (to drop NAs)
pca_df_psych <- pca_df %>%
  drop_na(c("id_row", "conflpplh", "cntrlife", "stflifer"))

# Calculate a manual reference Risk score to know the direction of risk (and correct the PCA sign if ne

risk_ref_psych <- pca_df_psych %>%
  mutate(across(
    c("conflpplh", "cntrlife", "stflifer"),
    ~ (.-min(., na.rm=TRUE)) / (max(., na.rm=TRUE) - min(., na.rm=TRUE))
  )) %>%
  transmute(risk_ref_psych = rowMeans(across(c("conflpplh", "cntrlife", "stflifer")), na.rm = TRUE)) %>%
  pull()

# Step 1: Run PCA
pca_psych <- prcomp(pca_df_psych[, c("conflpplh", "cntrlife", "stflifer")],
  scale. = TRUE)

print(pca_psych)

## Standard deviations (1, .., p=3):
## [1] 1.2175119 0.9636095 0.7675426
##
## Rotation (n x k) = (3 x 3):
##          PC1      PC2      PC3
## conflpplh 0.3599847 0.9324708 -0.03015402
## cntrlife  0.6565221 -0.2761526 -0.70193909
## stflifer  0.6628648 -0.2328906 0.71159838

#Check if sign is aligned with risk reference
cor_psych <- cor(pca_psych$x[,1], risk_ref_psych, use = "complete.obs")

if(cor_psych < 0){
  pca_psych$x[,1] <- -pca_psych$x[,1]
  pca_psych$rotation[,1] <- -pca_psych$rotation[,1]
}

```

```

print(pca_psych)

## Standard deviations (1, ..., p=3):
## [1] 1.2175119 0.9636095 0.7675426
##
## Rotation (n x k) = (3 x 3):
##           PC1      PC2      PC3
## conflpplh 0.3599847 0.9324708 -0.03015402
## cntrlife  0.6565221 -0.2761526 -0.70193909
## stflifer  0.6628648 -0.2328906 0.71159838

# Merge the result of the PCA to the initial pca_df

pca_df_psych_scores <- pca_df_psych %>%
  mutate(pc1_psych = pca_psych$x[,1])

# Step 2: Define survey design
pca_design_psych <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~pspwght,
  data = pca_df_psych_scores,
  nest = TRUE
)

# Step 3: Estimate means with confidence intervals

#By sex
psych_pca_bysex <- svyby(
  ~pc1_psych,
  ~cntry + sex,
  pca_design_psych,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
)

#pooled sex
psych_pca_pooledsex <- svyby(
  ~pc1_psych,
  ~cntry,
  pca_design_psych,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
) %>%
  as.data.frame() %>%
  mutate(sex = "All") # add label so we can row-bind

# Europe pooled (mean of countries)

psych_pca_pooledCountry <- mean(psych_pca_pooledsex$pc1_psych, na.rm = TRUE)

```



```

# WR pooled (mean of countries)

psych_pca_pooledWR <- psych_pca_pooledsex %>%
  left_join(country_regionWelfare, by = "cntry") %>%
  group_by(regionWelfare) %>%
  summarise(
    wr_mean_psych = mean(pc1_psych, na.rm = TRUE),
    .groups = "drop"
  )

# Step 4: Bind analyses, and add region info back in (once per country)

psych_pca_score <- bind_rows(
  psych_pca_pooledsex %>% dplyr::select(cntry, pc1_psych, se, ci_l, ci_u, sex),
  psych_pca_bysex %>% dplyr::select(cntry, pc1_psych, se, ci_l, ci_u, sex)
)

psych_pca_score <- psych_pca_score %>%
  left_join(country_regionWelfare, by = "cntry")

```

Social Capital

```

# Step 0: Make a df only for the subset of variables (to drop NAs)
pca_df_sc <- pca_df %>%
  drop_na(c("id_row", "socialmeet", "inperdisc", "ppltrust", "trustplt", "volunfpr"))

# Calculate a manual reference Risk score to know the direction of risk (and correct the PCA sign if ne

risk_ref_sc <- pca_df_sc %>%
  mutate(across(
    c("socialmeet", "inperdisc", "ppltrust", "trustplt", "volunfpr"),
    ~ (.-min(., na.rm=TRUE)) / (max(., na.rm=TRUE) - min(., na.rm=TRUE))
  )) %>%
  transmute(risk_ref_sc = rowMeans(across(c("socialmeet", "inperdisc", "ppltrust", "trustplt", "volunfpr"),
    pull()

# Step 1: Run PCA

pca_sc <- prcomp(pca_df_sc[, c("socialmeet", "inperdisc", "ppltrust", "trustplt", "volunfpr")],
  scale. = TRUE) # Run PCA on selected variables

print(pca_sc)

## Standard deviations (1, .., p=5):
## [1] 1.3147888 1.0202514 0.9452431 0.8305819 0.8044045
##
## Rotation (n x k) = (5 x 5):
##
##          PC1      PC2      PC3      PC4      PC5
## socialmeet -0.3766034  0.6291543 -0.34237153  0.58698393  0.02379892

```

```

## inperdisc -0.5148603 0.2859868 -0.19503643 -0.75867943 0.19873789
## ppltrust -0.5166722 -0.3900564 -0.13065902 0.04077660 -0.74978086
## trustplt -0.4529631 -0.5689220 -0.04240319 0.26889742 0.63011771
## volunfpr -0.3477974 0.2157781 0.90877540 0.07672567 -0.02678047

# Check if sign is aligned with risk reference
cor_sc <- cor(pca_sc$x[,1], risk_ref_sc, use = "complete.obs")

if(cor_sc < 0){
  pca_sc$x[,1] <- -pca_sc$x[,1]
  pca_sc$rotation[,1] <- -pca_sc$rotation[,1]
}
print(pca_sc)

## Standard deviations (1, ..., p=5):
## [1] 1.3147888 1.0202514 0.9452431 0.8305819 0.8044045
##
## Rotation (n x k) = (5 x 5):
##          PC1      PC2      PC3      PC4      PC5
## socialmeet 0.3766034 0.6291543 -0.34237153 0.58698393 0.02379892
## inperdisc 0.5148603 0.2859868 -0.19503643 -0.75867943 0.19873789
## ppltrust 0.5166722 -0.3900564 -0.13065902 0.04077660 -0.74978086
## trustplt 0.4529631 -0.5689220 -0.04240319 0.26889742 0.63011771
## volunfpr 0.3477974 0.2157781 0.90877540 0.07672567 -0.02678047

# Merge pca_df_hc_scores to have the individual PC
pca_df_sc_scores <- pca_df_sc %>%
  mutate(pc1_sc = pca_sc$x[,1])

# Step 2: Define survey design
pca_design_sc <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~pspwght,
  data = pca_df_sc_scores,
  nest = TRUE
)

# Step 3: Estimate means with confidence intervals

#By sex
sc_pca_bysex <- svyby(
  ~pc1_sc,
  ~cstry + sex,
  pca_design_sc,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
)

#pooled sex
sc_pca_pooledsex <- svyby(
  ~pc1_sc,

```

```

  ~cntry,
  pca_design_sc,
  svymean,
  na.rm = TRUE,
  vartype = c("se", "ci")
) %>%
  as.data.frame() %>%
  mutate(sex = "All") # add label so we can row-bind

# Europe pooled (mean of countries)

sc_pca_pooledCountry <- mean(sc_pca_pooledsex$pc1_sc, na.rm = TRUE)

# WR pooled (mean of countries)

sc_pca_pooledWR <- sc_pca_pooledsex %>%
  left_join(country_regionWelfare, by = "cntry") %>%
  group_by(regionWelfare) %>%
  summarise(
    wr_mean_sc = mean(pc1_sc, na.rm = TRUE),
    .groups = "drop"
  )

# Step 4: Bind analyses, and add region info back in (once per country)

sc_pca_score <- bind_rows(
  sc_pca_pooledsex %>% dplyr::select(cntry, pc1_sc, se, ci_l, ci_u, sex),
  sc_pca_bysex %>% dplyr::select(cntry, pc1_sc, se, ci_l, ci_u, sex)
)

sc_pca_score <- sc_pca_score %>%
  left_join(country_regionWelfare, by = "cntry")

```

PLOTS: PCA Composite Scores (Supplementary material)

Education

```

# Get positions of countries per regime
regime_spans <- edu_index_score %>%
  arrange(regionWelfare, cntry) %>%
  distinct(cntry, regionWelfare) %>%
  mutate(cntry_id = as.numeric(factor(cntry, levels = unique(cntry)))) %>%
  group_by(regionWelfare) %>%
  summarise(x_start = min(cntry_id), x_end = max(cntry_id), .groups = "drop")

# Get positions of the lines for regime averages
regime_spans_eduWR <- regime_spans %>%
  left_join(edu_index_pooledWR, by = "regionWelfare")

#plot
ggplot(edu_index_score %>%
  arrange(regionWelfare) %>%

```

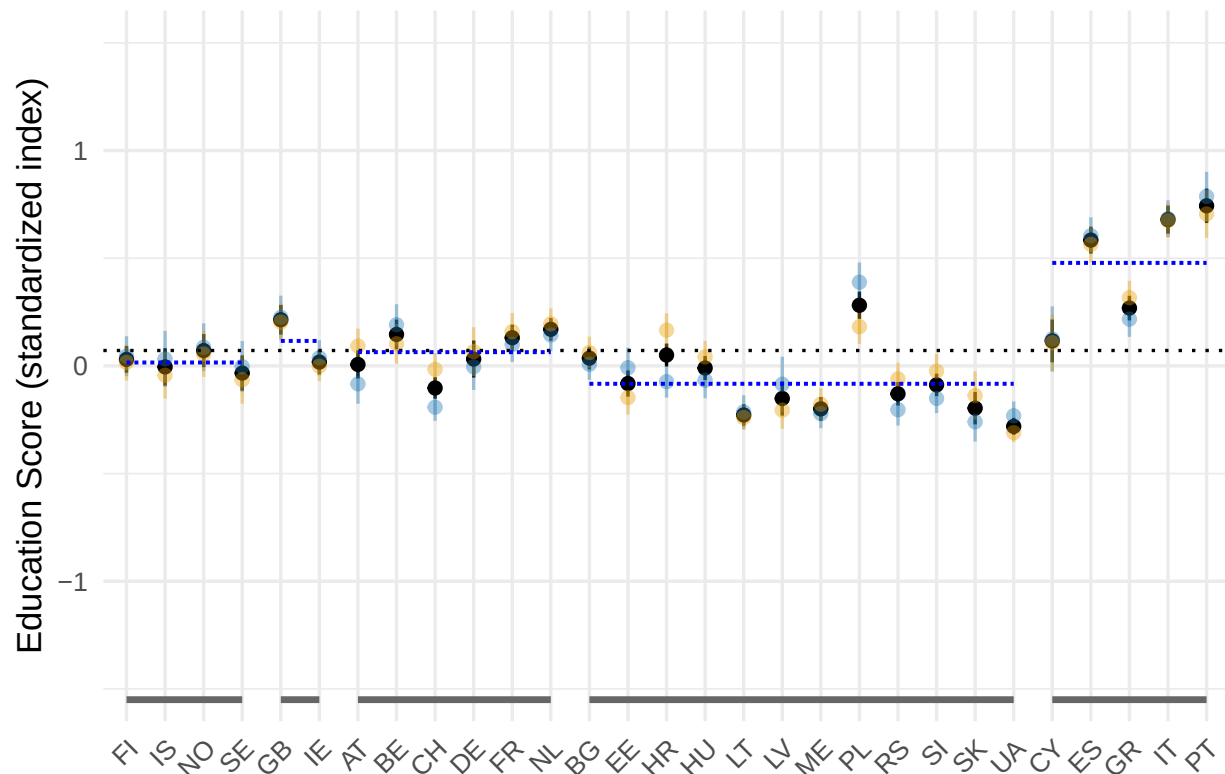
```

    mutate(cntry = factor(cntry, levels = unique(cntry))),
    aes(x = cntry, y = index_edu, ymin = ci_l, ymax = ci_u, color = sex)) +
geom_pointrange(size = 0.28,
    aes(alpha = ifelse(sex == "All", 1, 0.4)) ) +
scale_color_manual(
    values = c("All" = "black", "Men" = "#1F77B4", "Women" = "#E69F00"),
    breaks = c("All", "Men", "Women")
) +
scale_alpha_identity() +
geom_hline(yintercept = edu_index_pooledCountry, linetype = "dotted", color = "black") +
geom_segment(
    data = regime_spans_eduWR,
    aes(x = x_start, xend = x_end, y = wr_mean_edu, yend = wr_mean_edu),
    inherit.aes = F, linetype = "11", linewidth = 0.7, color = "blue") +
geom_segment(
    data = regime_spans,
    aes(x = x_start, xend = x_end, y = -1.55, yend = -1.55),
    inherit.aes = FALSE, linewidth = 1.2, color = "grey40"
) +

coord_cartesian(ylim = c(-1.5, 1.5)) +
theme_minimal(base_size = 13) +
labs(
    title = "Education disadvantage",
    y = "Education Score (standardized index)"
) +
theme(axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title.x = element_blank(),
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA),
    legend.position = "none")

```

Education disadvantage



```
ggsave("Figures\\CompositeScores\\eduWR_psp.png", width = 6, height = 4, dpi = 300)
```

Employment Conditions

```
# Get positions of countries per regime
regime_spans <- work_pca_score %>%
  arrange(regionWelfare, cntry) %>%
  distinct(cntry, regionWelfare) %>%
  mutate(cntry_id = as.numeric(factor(cntry, levels = unique(cntry)))) %>%
  group_by(regionWelfare) %>%
  summarise(x_start = min(cntry_id), x_end = max(cntry_id), .groups = "drop")

# Get positions of the lines for regime averages
regime_spans_workWR <- regime_spans %>%
  left_join(work_pca_pooledWR, by = "regionWelfare")

#plot

ggplot(work_pca_score %>%
  arrange(regionWelfare) %>%
  mutate(cntry = factor(cntry, levels = unique(cntry))),
  aes(x = cntry, y = pc1_work, ymin = ci_l, ymax = ci_u, color = sex)) +
  geom_pointrange(size = 0.28,
    aes(alpha = ifelse(sex == "All", 1, 0.4)) ) +
  scale_color_manual(
```

```

  values = c("All" = "black", "Men" = "#1F77B4", "Women" = "#E69F00"),
  breaks = c("All", "Men", "Women")
) +
scale_alpha_identity() +
geom_hline(yintercept = work_pca_pooledCountry, linetype = "dotted", color = "black") +
geom_segment(
  data = regime_spans_workWR,
  aes(x = x_start, xend = x_end, y = wr_mean_work, yend = wr_mean_work),
  inherit.aes = F, linetype = "11", linewidth = 0.7, color = "blue") +
geom_segment(
  data = regime_spans,
  aes(x = x_start, xend = x_end, y = -1.55, yend = -1.55),
  inherit.aes = FALSE,
  size = 1.2,
  color = "grey40"
) +
coord_cartesian(ylim = c(-2, 2)) +
theme_minimal(base_size = 13) +
labs(
  title = "Employment disadvantage",
  y = "Composite Employment Score (PCA)"
) +
theme(axis.text.x = element_text(angle = 45, hjust = 1),
      axis.title.x = element_blank(),
      plot.background = element_rect(fill = "white", color = NA),
      panel.background = element_rect(fill = "white", color = NA),
      legend.position = "none")

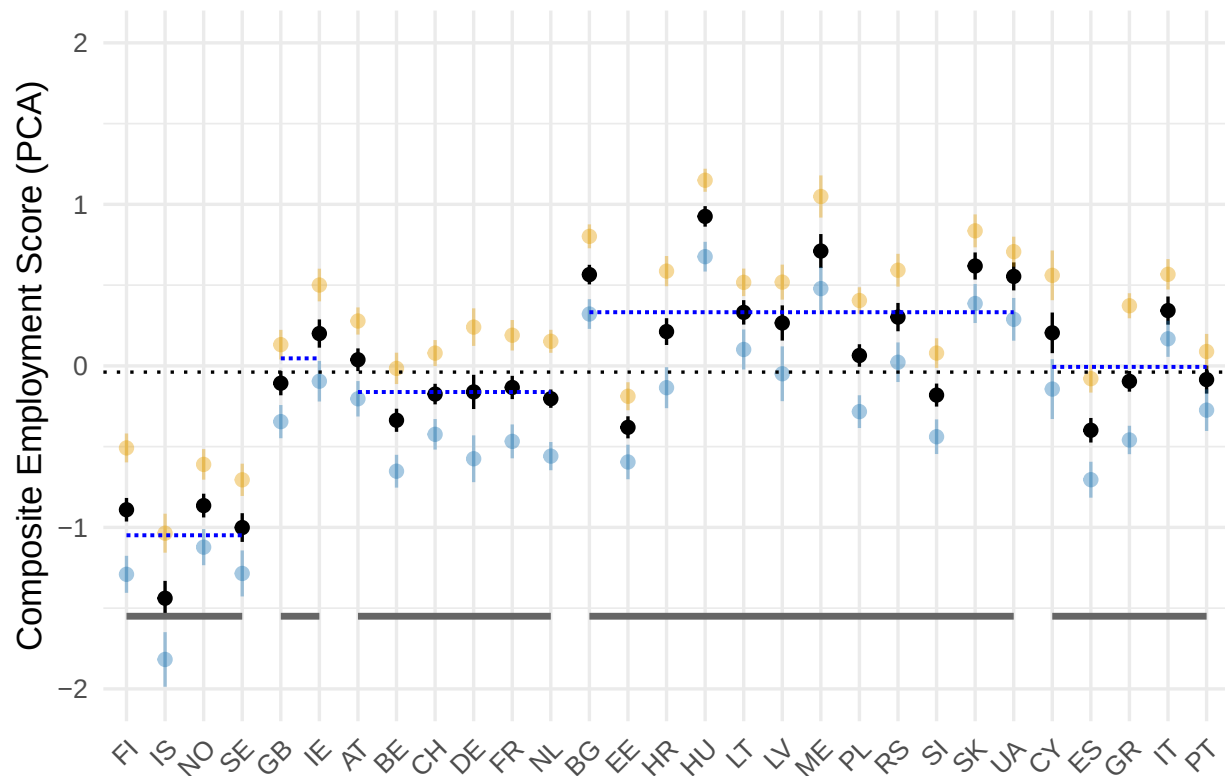
```

```

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

Employment disadvantage



```
ggsave("Figures\\CompositeScores\\workWR_psp.png", width = 6, height = 4, dpi = 300)
```

Material

```
# Get positions of countries per regime
regime_spans <- mat_pca_score %>%
  arrange(regionWelfare, cntry) %>%
  distinct(cntry, regionWelfare) %>%
  mutate(cntry_id = as.numeric(factor(cntry, levels = unique(cntry)))) %>%
  group_by(regionWelfare) %>%
  summarise(x_start = min(cntry_id), x_end = max(cntry_id), .groups = "drop")

# Get positions of the lines for regime averages
regime_spans_matWR <- regime_spans %>%
  left_join(mat_pca_pooledWR, by = "regionWelfare")

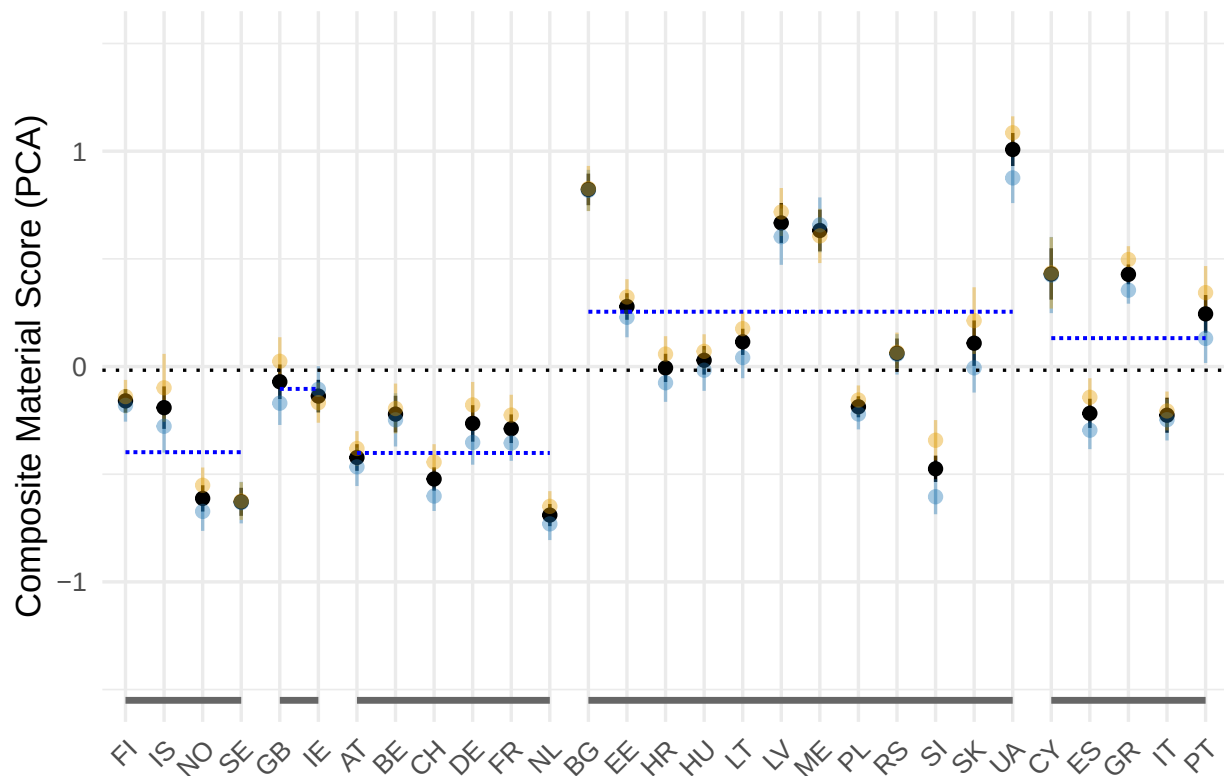
#plot
ggplot(mat_pca_score %>%
  arrange(regionWelfare) %>%
  mutate(cntry = factor(cntry, levels = unique(cntry))),
  aes(x = cntry, y = pc1_mat, ymin = ci_l, ymax = ci_u, color = sex)) +
  geom_pointrange(size = 0.28,
    aes(alpha = ifelse(sex == "All", 1, 0.4)) ) +
  scale_color_manual(
    values = c("All" = "black", "Men" = "#1F77B4", "Women" = "#E69F00"),
    breaks = c("All", "Men", "Women")
```

```

) +
scale_alpha_identity() +
geom_hline(yintercept = mat_pca_pooledCountry, linetype = "dotted", color = "black") +
geom_segment(
  data = regime_spans_matWR,
  aes(x = x_start, xend = x_end, y = wr_mean_mat, yend = wr_mean_mat),
  inherit.aes = F, linetype = "11", linewidth = 0.7, color = "blue") +
geom_segment(
  data = regime_spans,
  aes(x = x_start, xend = x_end, y = -1.55, yend = -1.55),
  inherit.aes = FALSE,
  linewidth = 1.2,
  color = "grey40")
) +
coord_cartesian(ylim = c(-1.5, 1.5)) +
theme_minimal(base_size = 13) +
labs(
  title = "Comparison of Material risk",
  y = "Composite Material Score (PCA)"
) +
theme(axis.text.x = element_text(angle = 45, hjust = 1),
      axis.title.x = element_blank(),
      plot.background = element_rect(fill = "white", color = NA),
      panel.background = element_rect(fill = "white", color = NA),
      legend.position = "none")

```

Comparison of Material risk




```
ggsave("Figures\\CompositeScores\\matWR_psp.png", width = 6, height = 4, dpi = 300)
```

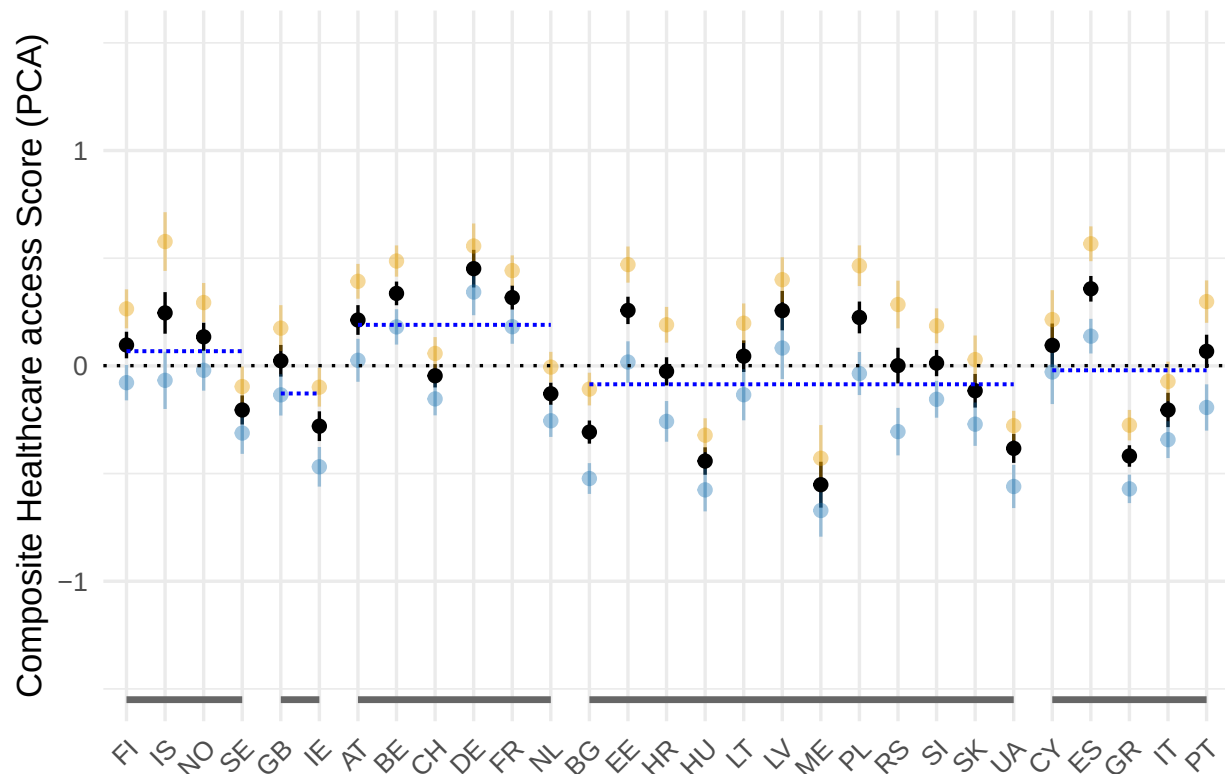
HC access

```
# Get positions of countries per regime
regime_spans <- hc_pca_score %>%
  arrange(regionWelfare, cntry) %>%
  distinct(cntry, regionWelfare) %>%
  mutate(cntry_id = as.numeric(factor(cntry, levels = unique(cntry)))) %>%
  group_by(regionWelfare) %>%
  summarise(x_start = min(cntry_id), x_end = max(cntry_id), .groups = "drop")

# Get positions of the lines for regime averages
regime_spans_hcWR <- regime_spans %>%
  left_join(hc_pca_pooledWR, by = "regionWelfare")

#plot
ggplot(hc_pca_score %>%
  arrange(regionWelfare) %>%
  mutate(cntry = factor(cntry, levels = unique(cntry))),
  aes(x = cntry, y = pc1_hc, ymin = ci_l, ymax = ci_u, color = sex)) +
  geom_pointrange(size = 0.28,
    aes(alpha = ifelse(sex == "All", 1, 0.4)) ) +
  scale_color_manual(
    values = c("All" = "black", "Men" = "#1F77B4", "Women" = "#E69F00"),
    breaks = c("All", "Men", "Women")
  ) +
  scale_alpha_identity() +
  geom_hline(yintercept = hc_pca_pooledCountry, linetype = "dotted", color = "black") +
  geom_segment(
    data = regime_spans_hcWR,
    aes(x = x_start, xend = x_end, y = wr_mean_hc, yend = wr_mean_hc),
    inherit.aes = F, linetype = "11", linewidth = 0.7, color = "blue" ) +
  geom_segment(
    data = regime_spans,
    aes(x = x_start, xend = x_end, y = -1.55, yend = -1.55),
    inherit.aes = FALSE,
    size = 1.2,
    color = "grey40"
  ) +
  coord_cartesian(ylim = c(-1.5, 1.5)) +
  theme_minimal(base_size = 13) +
  labs(
    title = "Healthcare access disadvantage",
    y = "Composite Healthcare access Score (PCA)"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title.x = element_blank(),
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA),
    legend.position = "none")
```

Healthcare access disadvantage



```
ggsave("Figures\\CompositeScores\\HCWR_psp.png", width = 6, height = 4, dpi = 300)
```

Behavioural

```
# Get positions of countries per regime
regime_spans <- behav_pca_score %>%
  arrange(regionWelfare, cntry) %>%
  distinct(cntry, regionWelfare) %>%
  mutate(cntry_id = as.numeric(factor(cntry, levels = unique(cntry)))) %>%
  group_by(regionWelfare) %>%
  summarise(x_start = min(cntry_id), x_end = max(cntry_id), .groups = "drop")

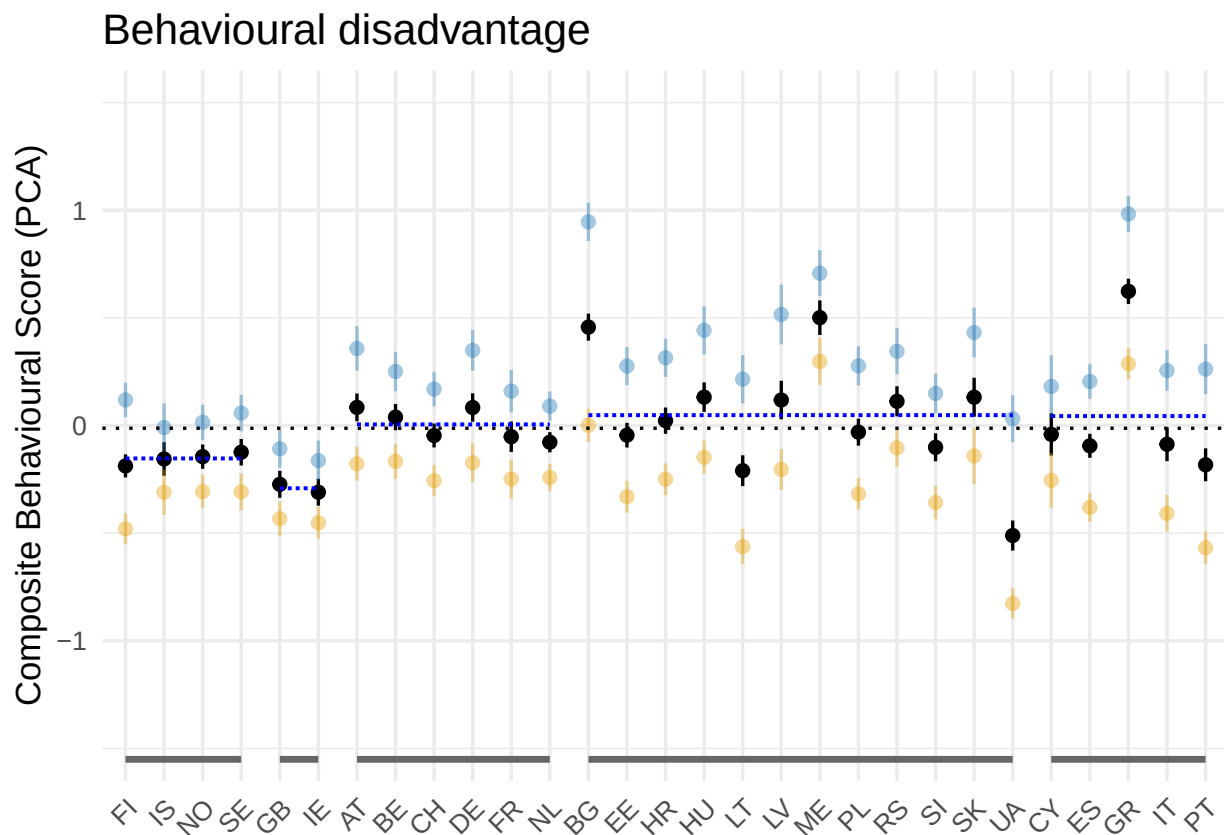
# Get positions of the lines for regime averages
regime_spans_behavWR <- regime_spans %>%
  left_join(behav_pca_pooledWR, by = "regionWelfare")

#plot
ggplot(behav_pca_score %>%
  arrange(regionWelfare) %>%
  mutate(cntry = factor(cntry, levels = unique(cntry))),
  aes(x = cntry, y = pc1_behav, ymin = ci_l, ymax = ci_u, color = sex)) +
  geom_pointrange(size = 0.28,
    aes(alpha = ifelse(sex == "All", 1, 0.4)) ) +
  scale_color_manual(
    values = c("All" = "black", "Men" = "#1F77B4", "Women" = "#E69F00"),
    breaks = c("All", "Men", "Women")
```

```

) +
scale_alpha_identity() +
geom_hline(yintercept = behav_pca_pooledCountry, linetype = "dotted", color = "black") +
geom_segment(
  data = regime_spans_behavWR,
  aes(x = x_start, xend = x_end, y = wr_mean_behav, yend = wr_mean_behav),
  inherit.aes = F, linetype = "11", linewidth = 0.7, color = "blue") +
geom_segment(
  data = regime_spans,
  aes(x = x_start, xend = x_end, y = -1.55, yend = -1.55),
  inherit.aes = FALSE,
  size = 1.2,
  color = "grey40")
) +
coord_cartesian(ylim = c(-1.5, 1.5)) +
theme_minimal(base_size = 13) +
labs(
  title = "Behavioural disadvantage",
  y = "Composite Behavioural Score (PCA)"
) +
theme(axis.text.x = element_text(angle = 45, hjust = 1),
      axis.title.x = element_blank(),
      plot.background = element_rect(fill = "white", color = NA),
      panel.background = element_rect(fill = "white", color = NA),
      legend.position = "none")

```



```
ggsave("Figures\\CompositeScores\\BehavWR_psp.png", width = 6, height = 4, dpi = 300)
```

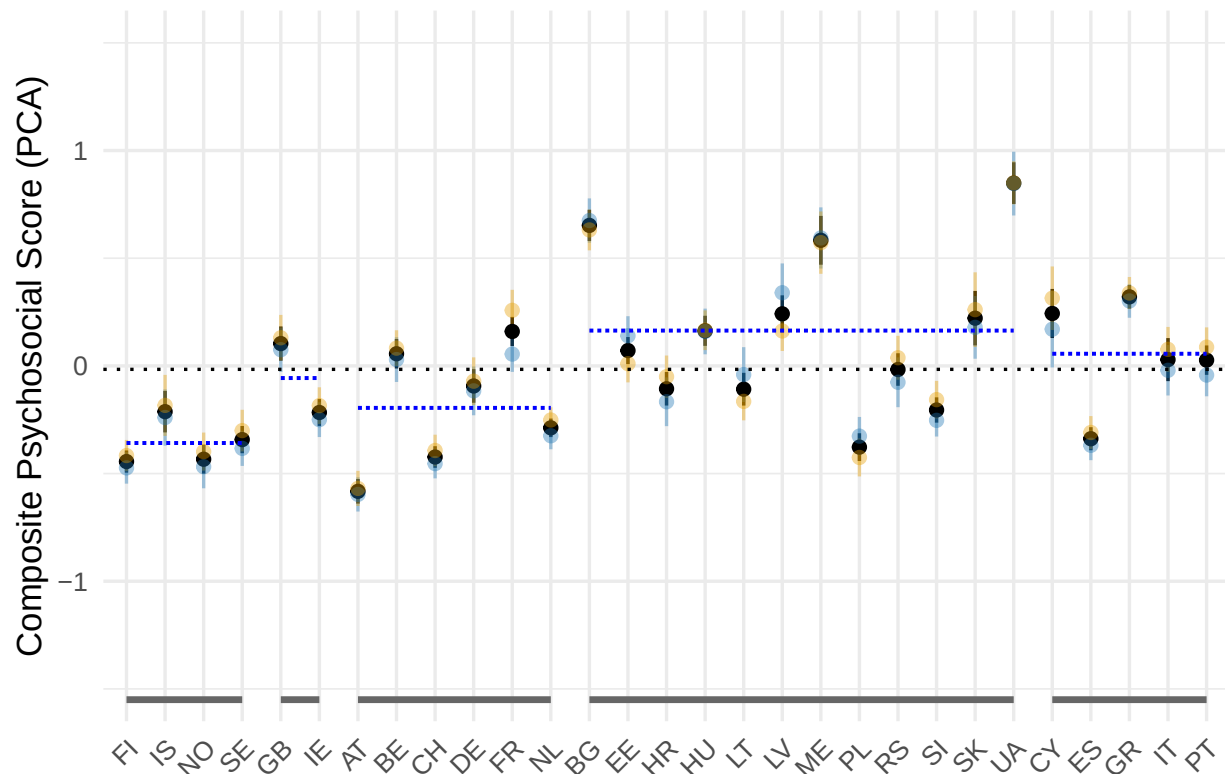
Psychosocial

```
# Get positions of countries per regime
regime_spans <- psych_pca_score %>%
  arrange(regionWelfare, cntry) %>%
  distinct(cntry, regionWelfare) %>%
  mutate(cntry_id = as.numeric(factor(cntry, levels = unique(cntry)))) %>%
  group_by(regionWelfare) %>%
  summarise(x_start = min(cntry_id), x_end = max(cntry_id), .groups = "drop")

# Get positions of the lines for regime averages
regime_spans_psychWR <- regime_spans %>%
  left_join(psych_pca_pooledWR, by = "regionWelfare")

#plot
ggplot(psych_pca_score %>%
  arrange(regionWelfare) %>%
  mutate(cntry = factor(cntry, levels = unique(cntry))),
  aes(x = cntry, y = pc1_psych, ymin = ci_l, ymax = ci_u, color = sex)) +
  geom_pointrange(size = 0.28,
    aes(alpha = ifelse(sex == "All", 1, 0.4)) ) +
  scale_color_manual(
    values = c("All" = "black", "Men" = "#1F77B4", "Women" = "#E69F00"),
    breaks = c("All", "Men", "Women")
  ) +
  scale_alpha_identity() +
  geom_hline(yintercept = psych_pca_pooledCountry, linetype = "dotted", color = "black") +
  geom_segment(
    data = regime_spans_psychWR,
    aes(x = x_start, xend = x_end, y = wr_mean_psych, yend = wr_mean_psych),
    inherit.aes = F, linetype = "11", linewidth = 0.7, color = "blue") +
  geom_segment(
    data = regime_spans,
    aes(x = x_start, xend = x_end, y = -1.55, yend = -1.55),
    inherit.aes = FALSE,
    size = 1.2,
    color = "grey40"
  ) +
  coord_cartesian(ylim = c(-1.5, 1.5)) +
  theme_minimal(base_size = 13) +
  labs(
    title = "Psychosocial disadvantage",
    y = "Composite Psychosocial Score (PCA)"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title.x = element_blank(),
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA),
    legend.position = "none")
```

Psychosocial disadvantage



```
ggsave("Figures\\CompositeScores\\psychWR_psp.png", width = 6, height = 4, dpi = 300)
```

Social Capital

```
# Get positions of countries per regime
regime_spans <- sc_pca_score %>%
  arrange(regionWelfare, cntry) %>%
  distinct(cntry, regionWelfare) %>%
  mutate(cntry_id = as.numeric(factor(cntry, levels = unique(cntry)))) %>%
  group_by(regionWelfare) %>%
  summarise(x_start = min(cntry_id), x_end = max(cntry_id), .groups = "drop")

# Get positions of the lines for regime averages
regime_spans_scWR <- regime_spans %>%
  left_join(sc_pca_pooledWR, by = "regionWelfare")

#plot

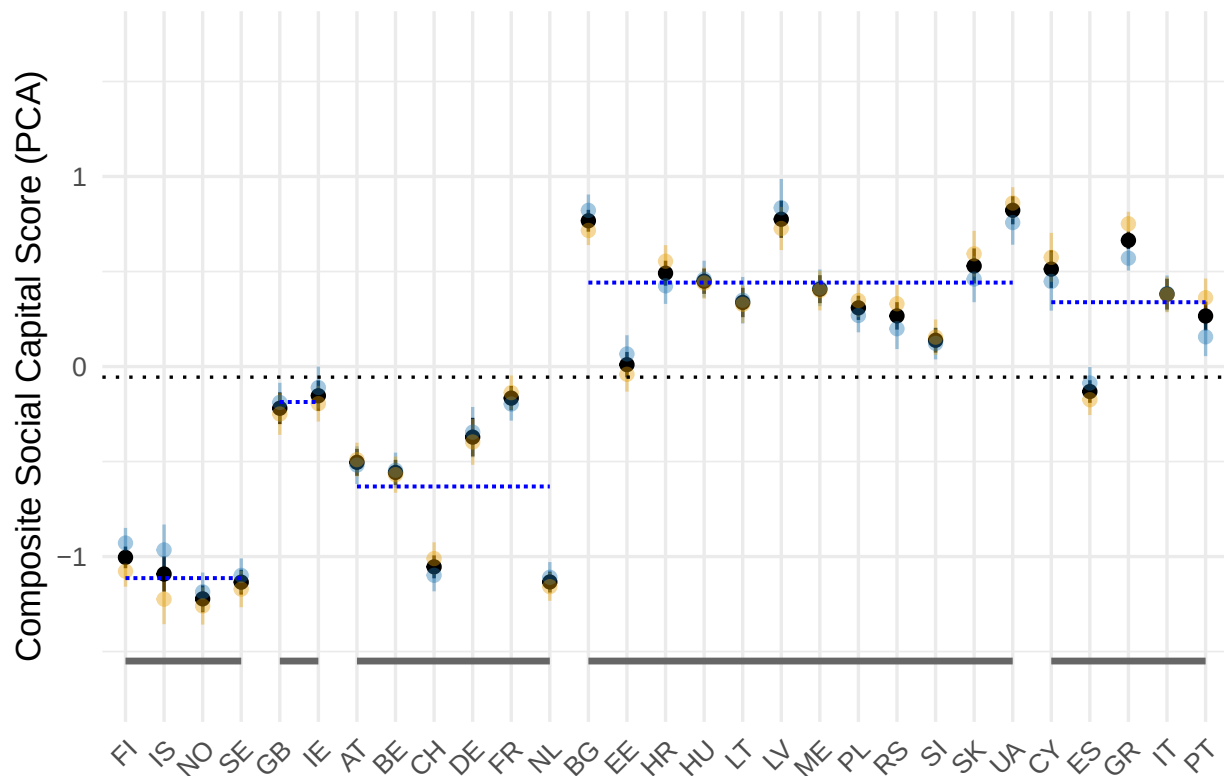
ggplot(sc_pca_score %>%
  arrange(regionWelfare) %>%
  mutate(cntry = factor(cntry, levels = unique(cntry))),
  aes(x = cntry, y = pc1_sc, ymin = ci_l, ymax = ci_u, color = sex)) +
  geom_pointrange(size = 0.28,
    aes(alpha = ifelse(sex == "All", 1, 0.4)) ) +
  scale_color_manual(
    values = c("All" = "black", "Men" = "#1F77B4", "Women" = "#E69F00"),
    breaks = c("All", "Men", "Women")
```

```

) +
scale_alpha_identity() +
geom_hline(yintercept = sc_pca_pooledCountry, linetype = "dotted", color = "black") +
geom_segment(
  data = regime_spans_scWR,
  aes(x = x_start, xend = x_end, y = wr_mean_sc, yend = wr_mean_sc),
  inherit.aes = F, linetype = "11", linewidth = 0.7, color = "blue") +
geom_segment(
  data = regime_spans,
  aes(x = x_start, xend = x_end, y = -1.55, yend = -1.55),
  inherit.aes = FALSE,
  size = 1.2,
  color = "grey40")
) +
coord_cartesian(ylim = c(-1.7, 1.7)) +
theme_minimal(base_size = 13) +
labs(
  title = "Social Capital disadvantage",
  y = "Composite Social Capital Score (PCA)"
) +
theme(axis.text.x = element_text(angle = 45, hjust = 1),
      axis.title.x = element_blank(),
      plot.background = element_rect(fill = "white", color = NA),
      panel.background = element_rect(fill = "white", color = NA),
      legend.position = "none")

```

Social Capital disadvantage



```
ggsave("Figures\\CompositeScores\\SCWR_psp.png", width = 6, height = 4, dpi = 300)
```

SPIDER-PLOTS: PCA

Prepare data

```
# Select relevant variables
```

```
pca_spider_vars <- c("index_edu", "pc1_work", "pc1_mat", "pc1_hc", "pc1_behav", "pc1_psych" , "pc1_sc")
```

```
# Rename variables for the graph
```

```
PCAblock_labels <- c(
  index_edu = "Education",
  pc1_work = "Employment",
  pc1_mat = "Material",
  pc1_hc = "HC access",
  pc1_behav = "Behaviour",
  pc1_psych = "Psychosocial",
  pc1_sc = "Social Capital"
)
```

```
### PCA data European average: prepare ----
```

```
all_pca_score_EuMean <- c(
  index_edu = edu_index_pooledCountry,
  pc1_work = work_pca_pooledCountry,
  pc1_mat = mat_pca_pooledCountry,
  pc1_hc = hc_pca_pooledCountry,
  pc1_behav = behav_pca_pooledCountry,
  pc1_psych = psych_pca_pooledCountry,
  pc1_sc = sc_pca_pooledCountry
)
```

```
### PCA data WR average ----
```

```
# merge allWR_pca_score into a new dataframe
```

```
allWR_pca_score <- merge(edu_index_pooledWR, work_pca_pooledWR, by = c("regionWelfare"))
allWR_pca_score <- merge(allWR_pca_score, mat_pca_pooledWR, by = c("regionWelfare"))
allWR_pca_score <- merge(allWR_pca_score, hc_pca_pooledWR, by = c("regionWelfare"))
allWR_pca_score <- merge(allWR_pca_score, behav_pca_pooledWR, by = c("regionWelfare"))
allWR_pca_score <- merge(allWR_pca_score, psych_pca_pooledWR, by = c("regionWelfare"))
allWR_pca_score <- merge(allWR_pca_score, sc_pca_pooledWR, by = c("regionWelfare"))
```

```
#convert all_pca_score to numeric
```

```
allWR_pca_numeric <- allWR_pca_score %>%
  dplyr::select(c(wr_mean_edu, wr_mean_work, wr_mean_mat, wr_mean_hc, wr_mean_behav, wr_mean_psych, wr_mean_sc),
  as.data.frame() %>%
```

```

lapply(function(x) {
  if (inherits(x, "haven_labelled")) {
    as.numeric(as_factor(x))
  } else {
    as.numeric(x)
  }
}) %>%
as.data.frame()

### PCA data Country: prepare ----

# clean_pca_score dataframes to only leave cntry and pca score

edu_index_tomerge <- edu_index_pooledsex %>%
  dplyr::select(cntry, index_edu)
work_pca_tomerge <- work_pca_pooledsex %>%
  dplyr::select(cntry, pc1_work)
mat_pca_tomerge <- mat_pca_pooledsex %>%
  dplyr::select(cntry, pc1_mat)
hc_pca_tomerge <- hc_pca_pooledsex %>%
  dplyr::select(cntry, pc1_hc)
behav_pca_tomerge <- behav_pca_pooledsex %>%
  dplyr::select(cntry, pc1_behav)
psych_pca_tomerge <- psych_pca_pooledsex %>%
  dplyr::select(cntry, pc1_psych)
sc_pca_tomerge <- sc_pca_pooledsex %>%
  dplyr::select(cntry, pc1_sc)

# merge all_pca_score into a new dataframe

all_pca_score <- merge(edu_index_tomerge, work_pca_tomerge, by = c("cntry"))
all_pca_score <- merge(all_pca_score, mat_pca_tomerge, by = c("cntry"))
all_pca_score <- merge(all_pca_score, hc_pca_tomerge, by = c("cntry"))
all_pca_score <- merge(all_pca_score, behav_pca_tomerge, by = c("cntry"))
all_pca_score <- merge(all_pca_score, psych_pca_tomerge, by = c("cntry"))
all_pca_score <- merge(all_pca_score, sc_pca_tomerge, by = c("cntry"))

#convert all_pca_score to numeric

all_pca_numeric <- all_pca_score %>%
  dplyr::select(all_of(pca_spider_vars)) %>%
  as.data.frame() %>%
  lapply(function(x) {
    if (inherits(x, "haven_labelled")) {
      as.numeric(as_factor(x))
    } else {
      as.numeric(x)
    }
  }) %>%
  as.data.frame()

```



```

### Max/Min for each variable ----
# Compute per-axis anchors (max and min) following order

pca_max_val <- sapply(all_pca_numeric[pca_spider_vars], max, na.rm = TRUE)
pca_min_val <- sapply(all_pca_numeric[pca_spider_vars], min, na.rm = TRUE)

```

Country plots

```

# Loop
for (country in country_list) {

  # Filter country-specific data
  country_data <- all_pca_score %>%
    filter(cntry == country) %>%
    dplyr::select(all_of(pca_spider_vars))%>%
    as.data.frame()

  # enforce the same column order explicitly
  country_data <- country_data[, pca_spider_vars, drop = FALSE]

  # construct radar df
  radarPCA_df <- rbind(
    pca_max_val[pca_spider_vars],
    pca_min_val[pca_spider_vars],
    country_data,
    all_pca_score_EuMean[pca_spider_vars]
  )

  colnames(radarPCA_df) <- PCAblock_labels[pca_spider_vars]
  par(lend = 2)

  # Showing plot
  par(mar = c(1, 2, 3, 2))

  radarchart(radarPCA_df,
    axistype = 1,

    pcol = c("darkred", "darkblue"),
    pfcol = c(scales::alpha("darkred", 0.2), NA),
    plwd = c(2, 2),
    cglcol = "grey",
    cglty = 1,
    axislabcol = "black",
    #caxislabels = rep("", length(caxislabels_seq)),
    cglwd = 0.8,
    vlce = 0.9,
    title = country)

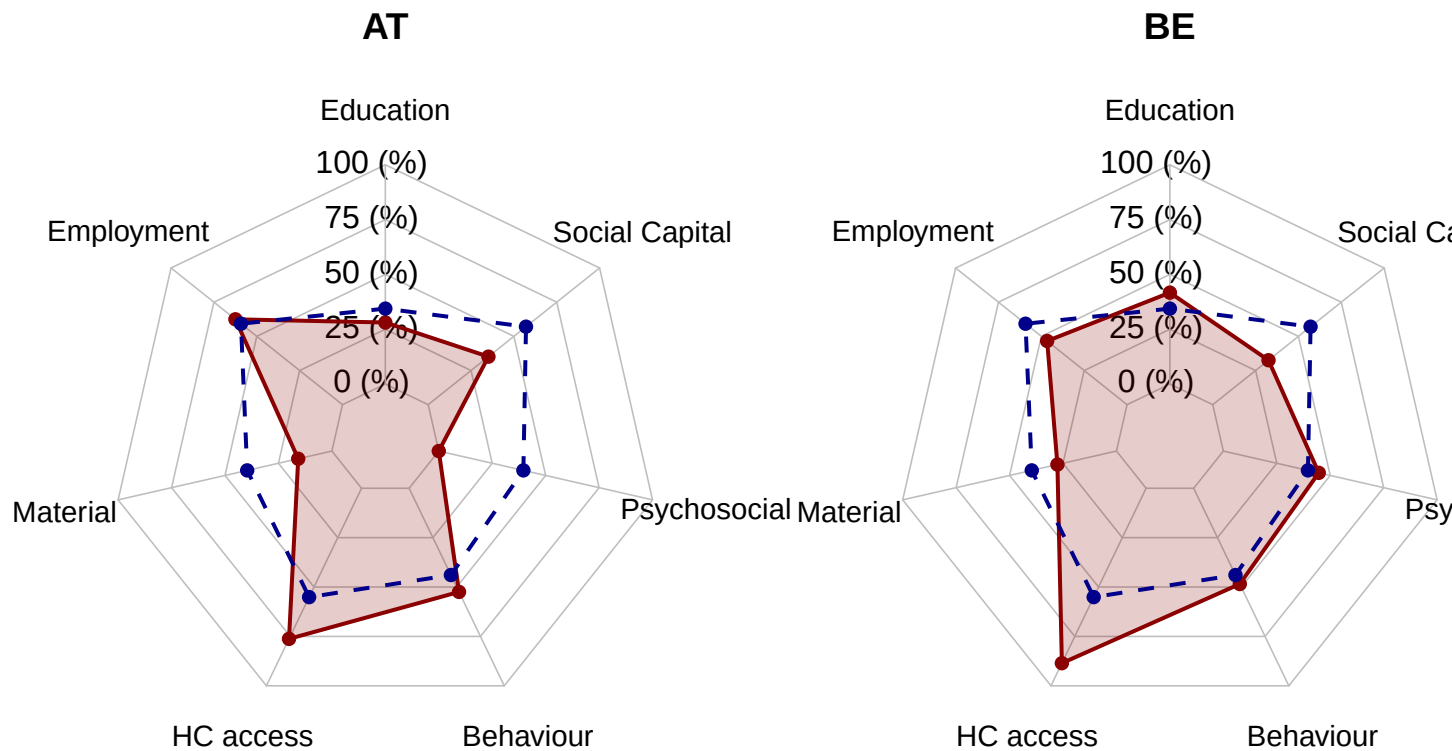
  Sys.sleep(1)
}

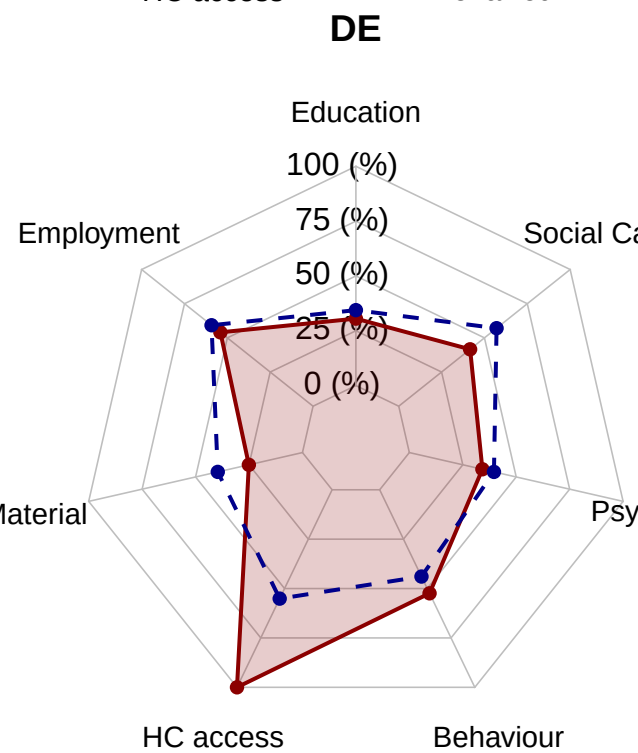
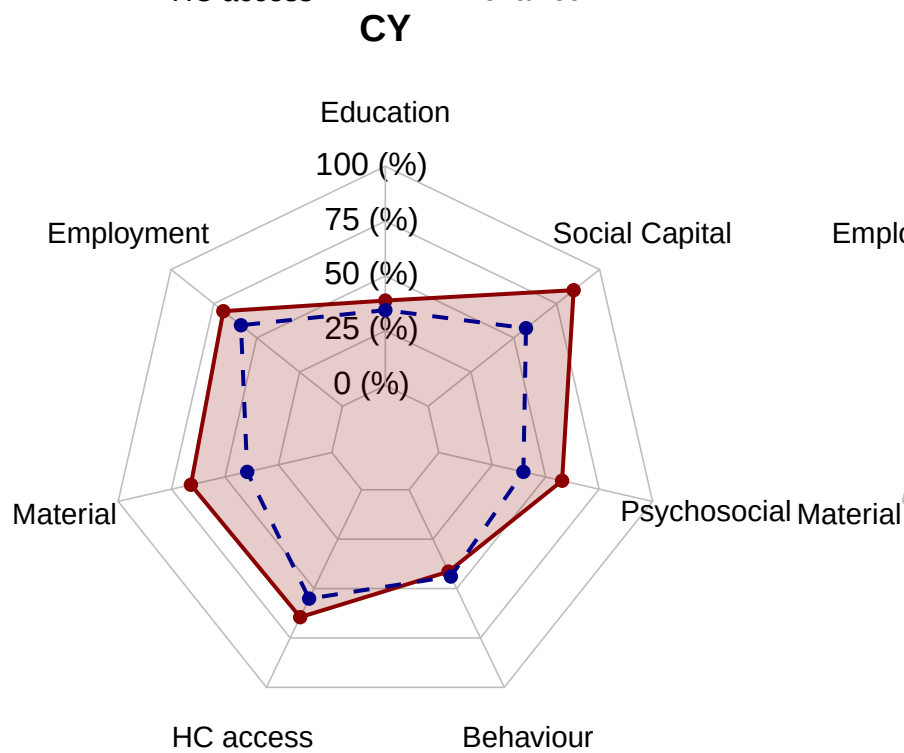
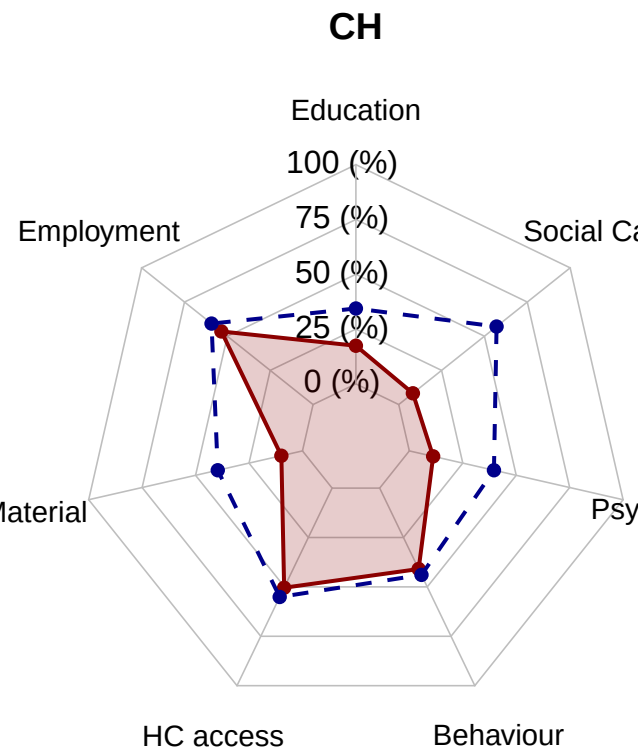
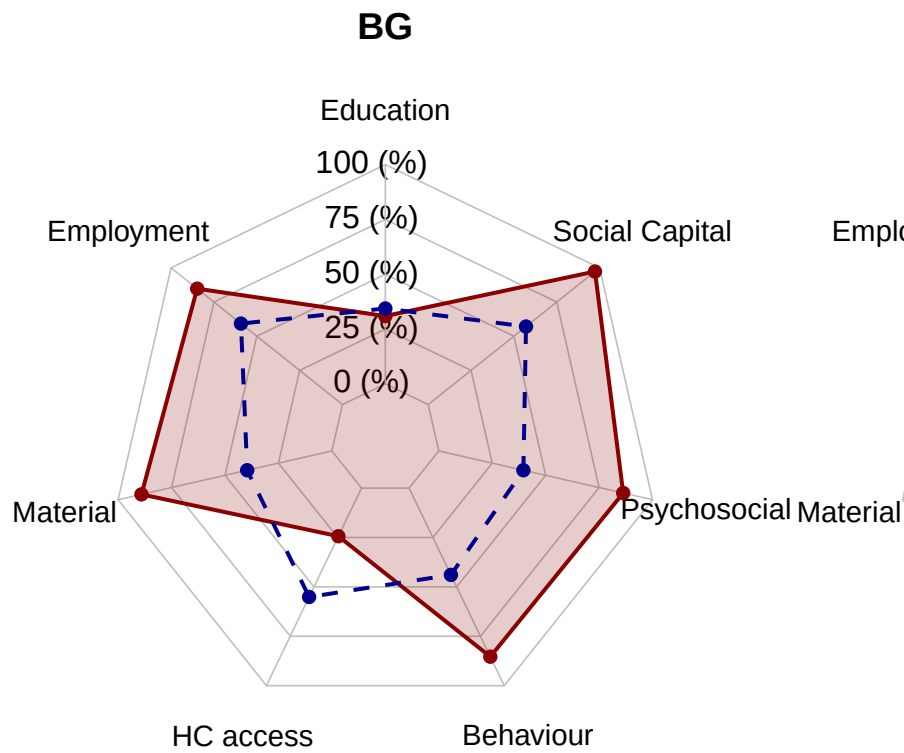
```

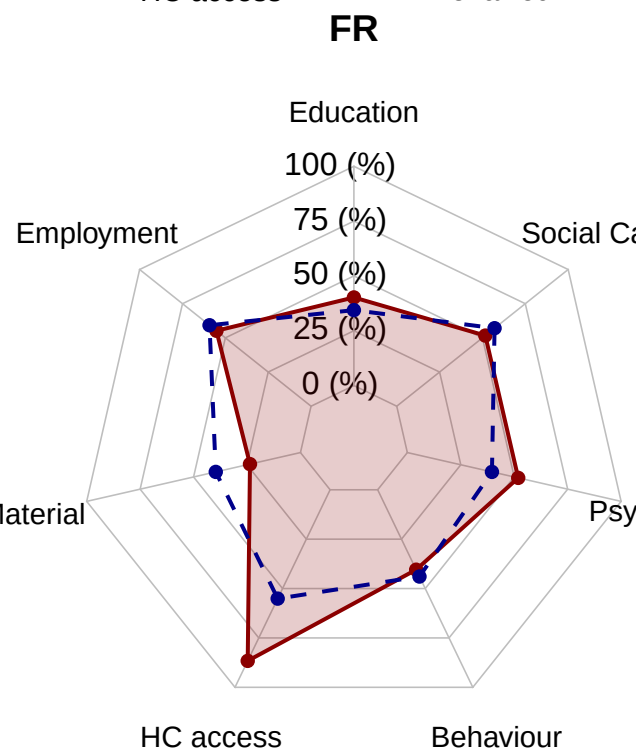
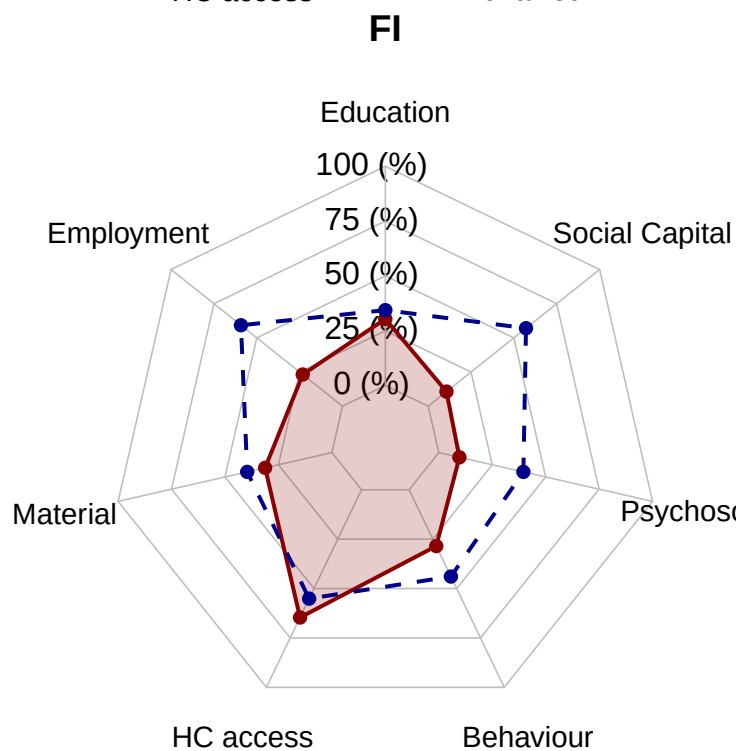
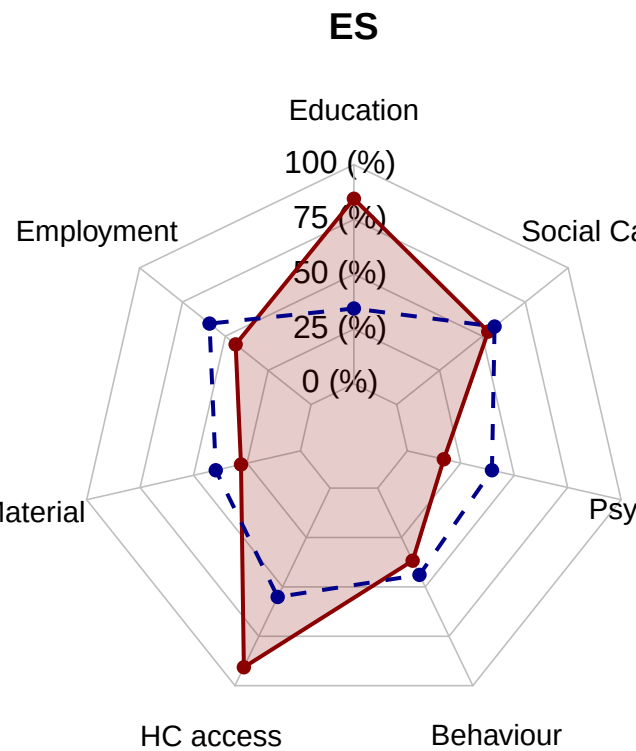
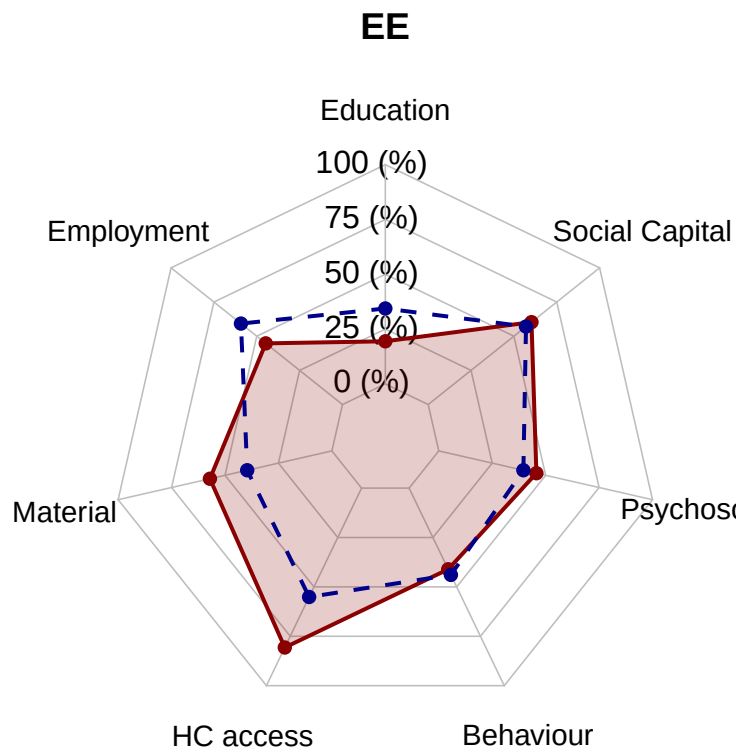
```

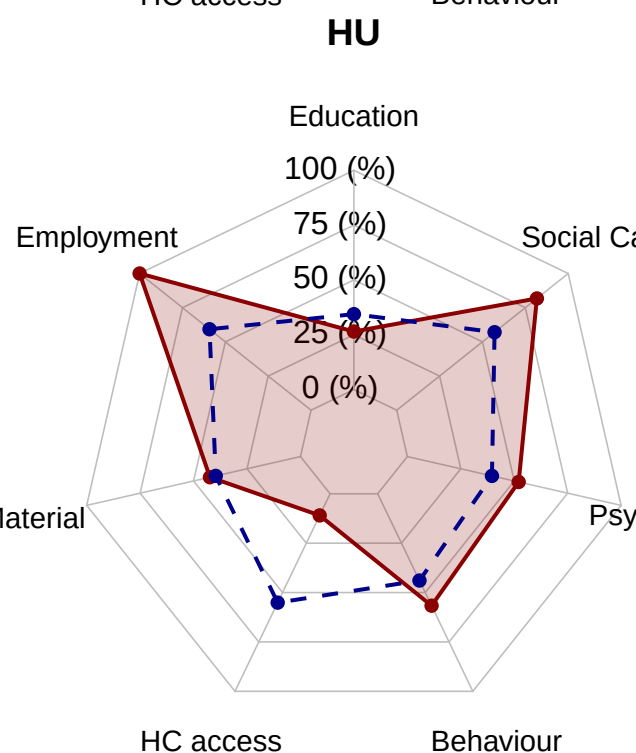
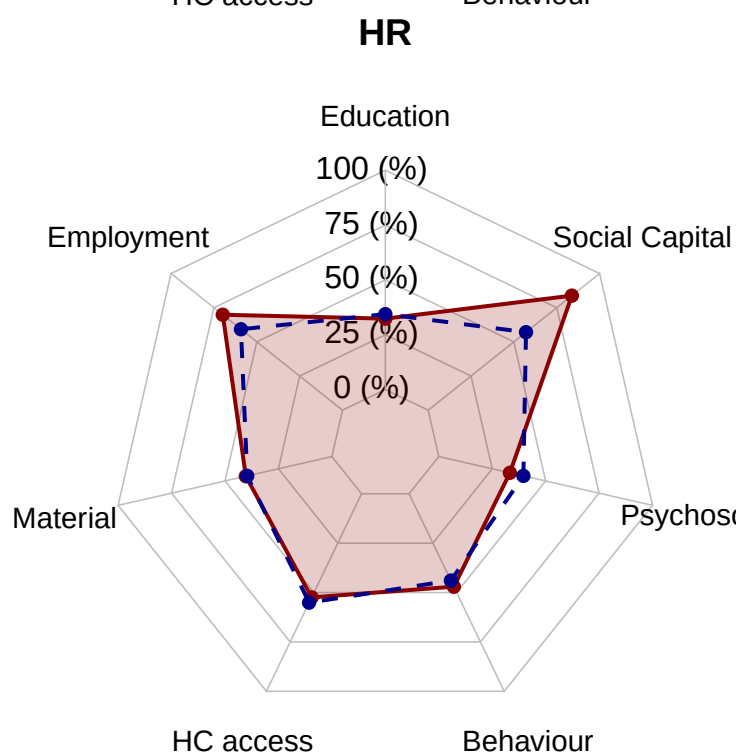
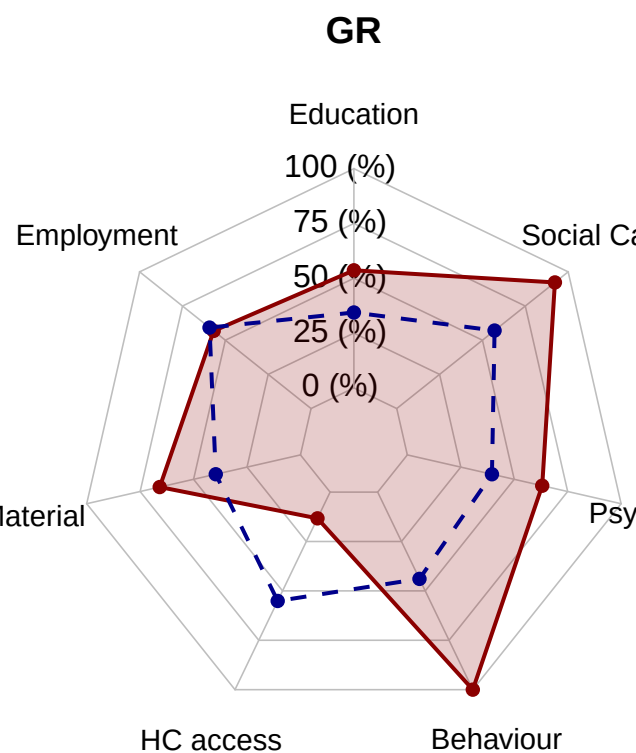
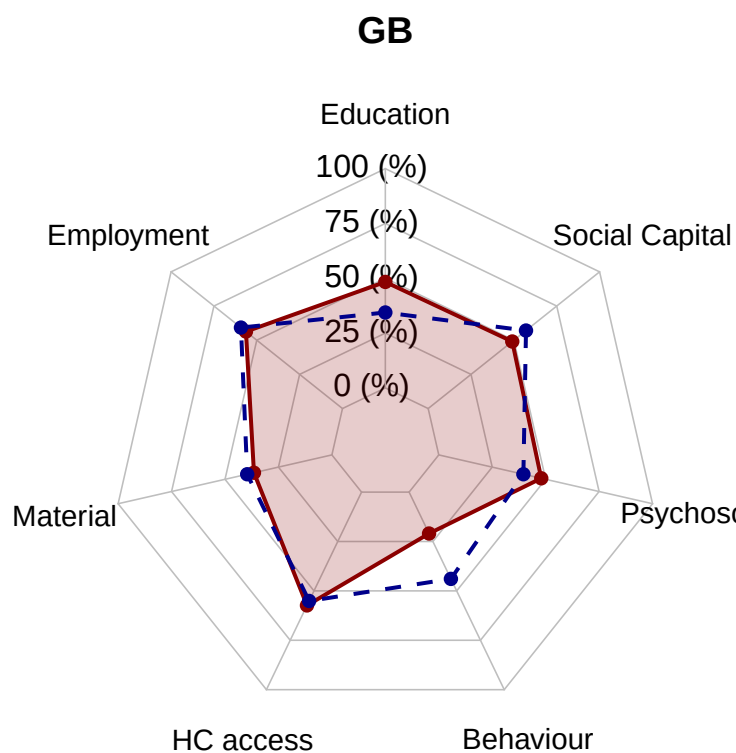
# Saving plot to PNG
png(filename = paste0("Figures\\pca_per_country_spider\\Spiderplot_", country, ".png"), width = 600, height = 600)
par(mar = c(1, 2, 3, 2))
radarchart(radarPCA_df,
  axistype = 1,
  pcol = c("darkred", "darkblue"),
  pfcol = c(scales::alpha("darkred", 0.2), NA),
  plwd = c(4, 4),
  cglcol = "grey",
  cglty = 1,
  axislabcol = "black",
  #caxislabels = rep("", length(caxislabels_seq)),
  cglwd = 0.8,
  vlceex = 0.9,
  title = country,
  cex.main = 4)
dev.off()
}

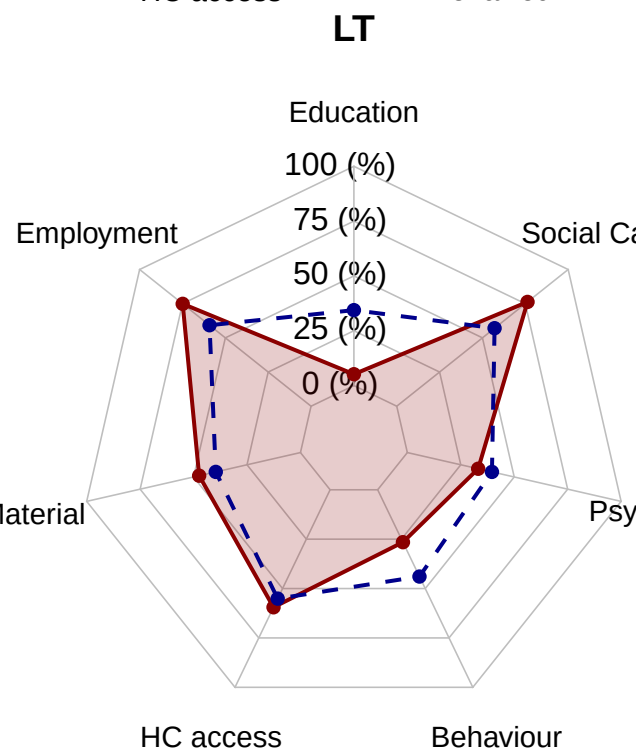
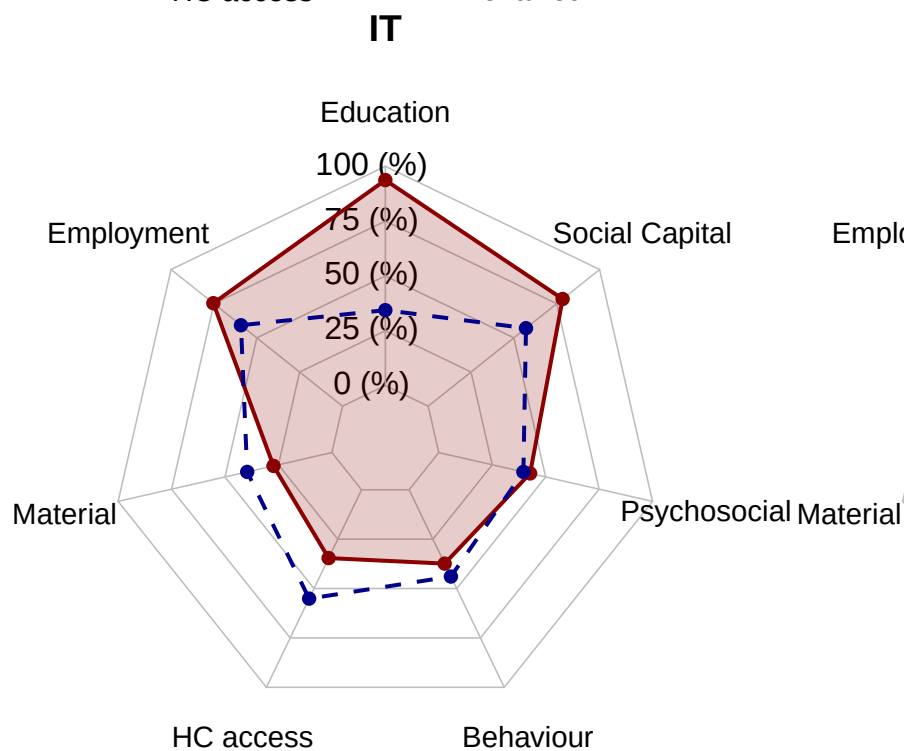
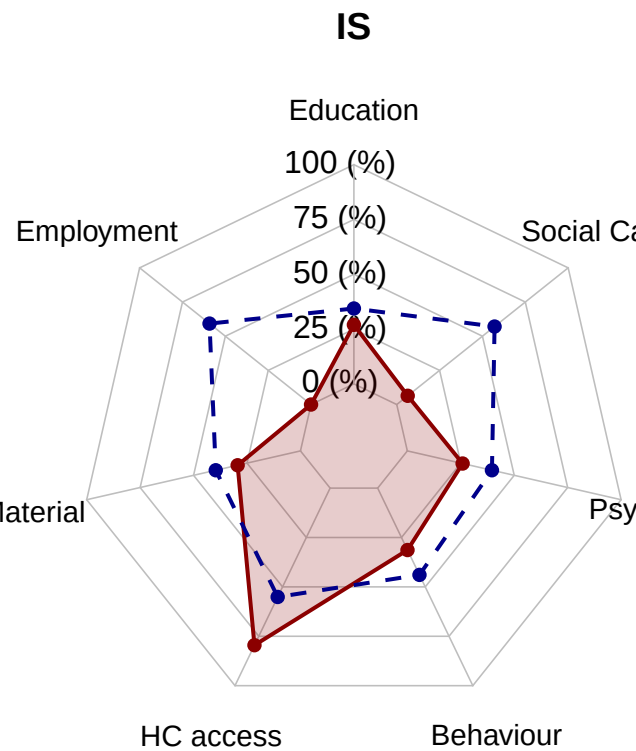
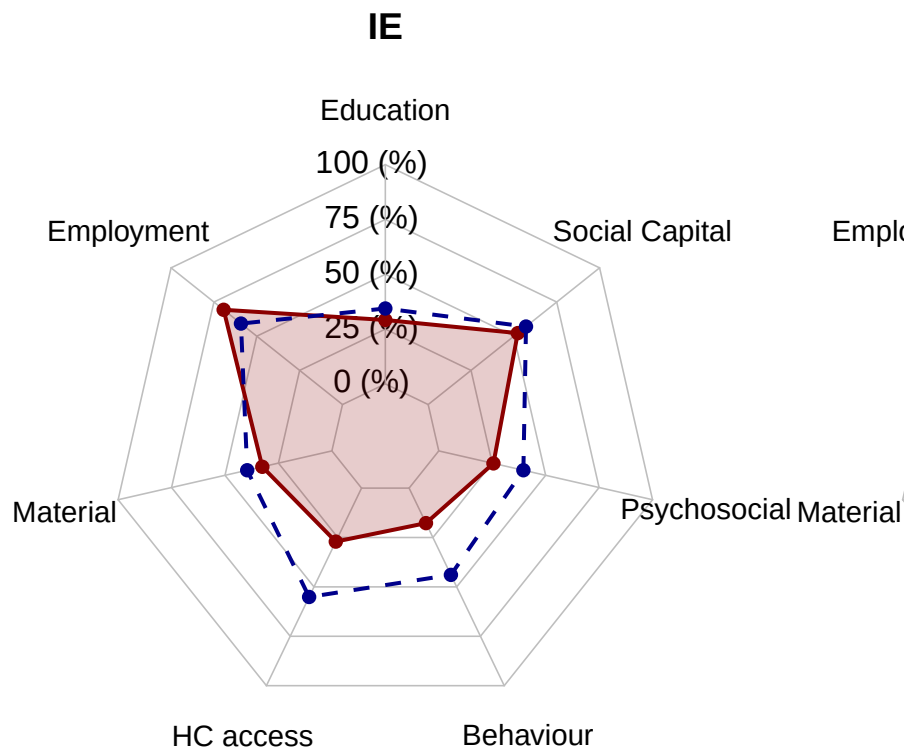
```

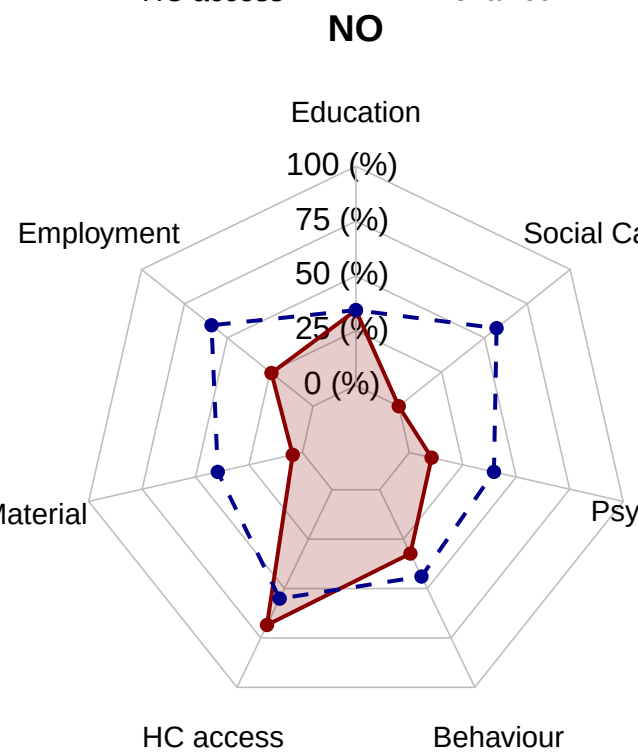
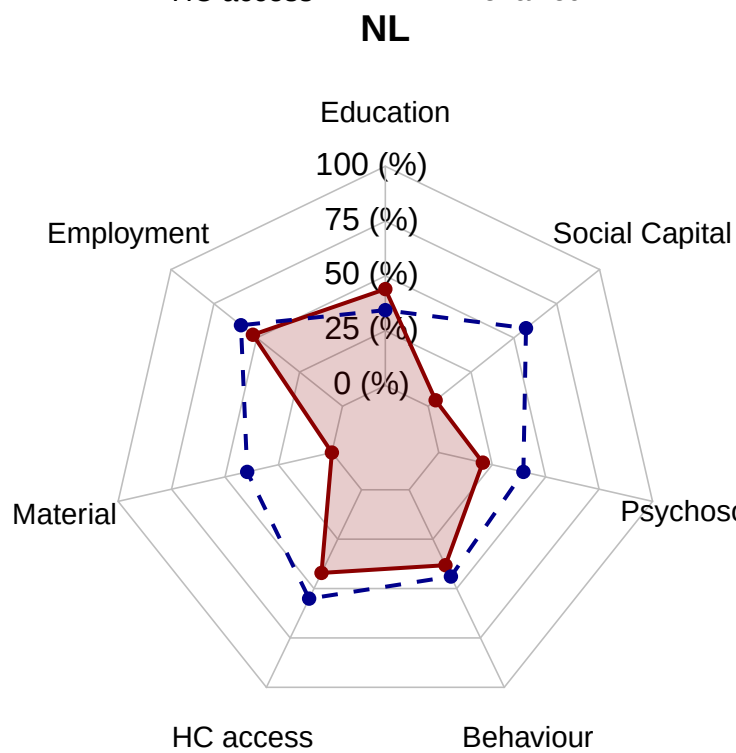
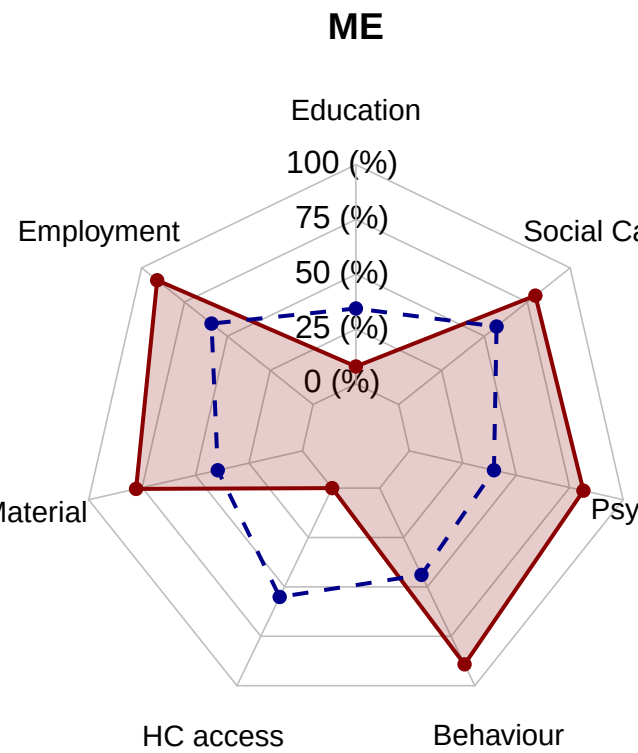
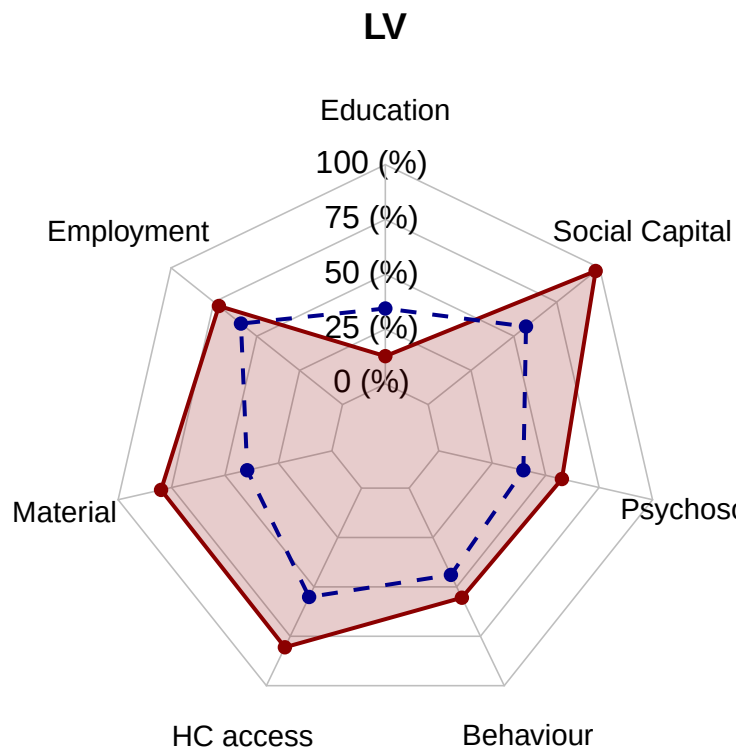


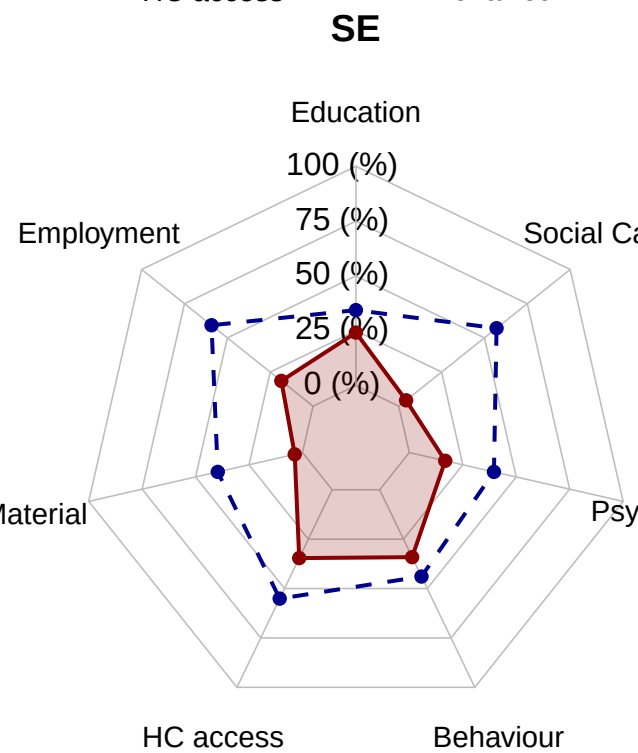
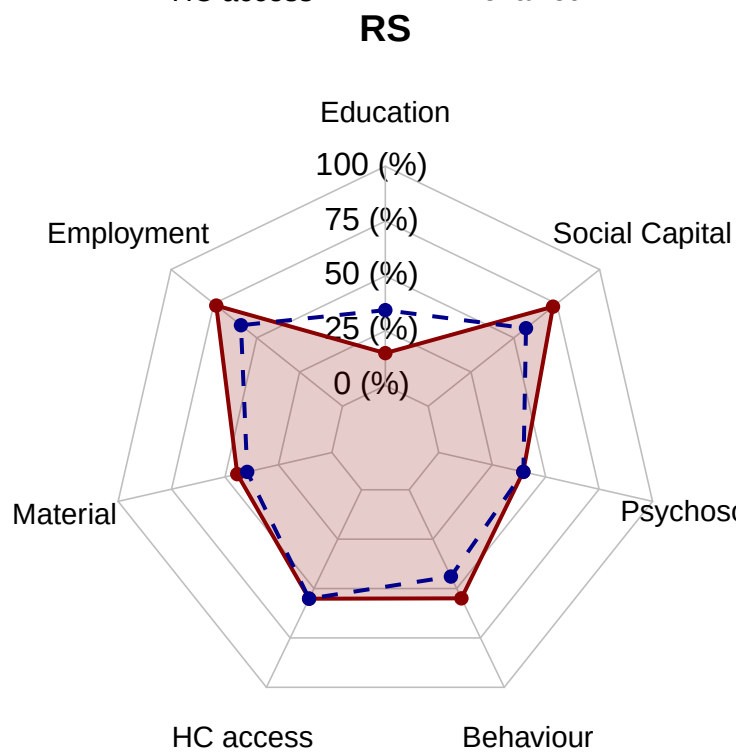
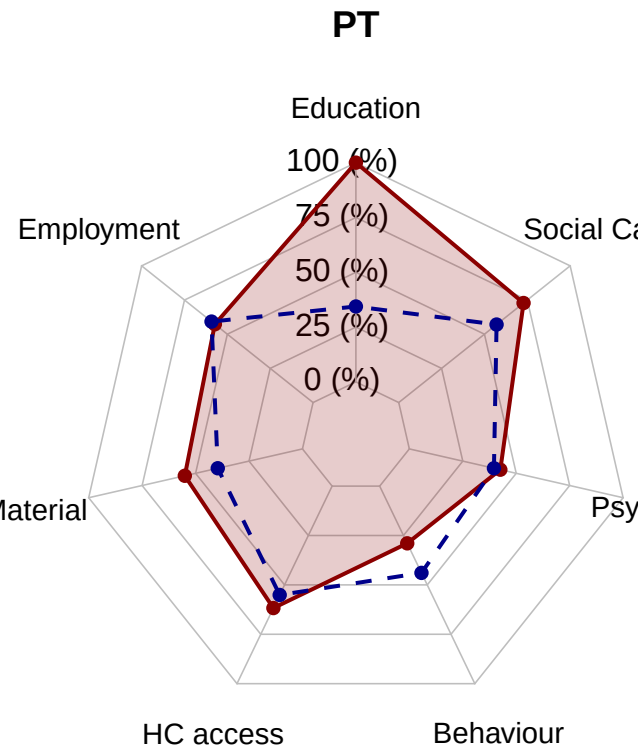
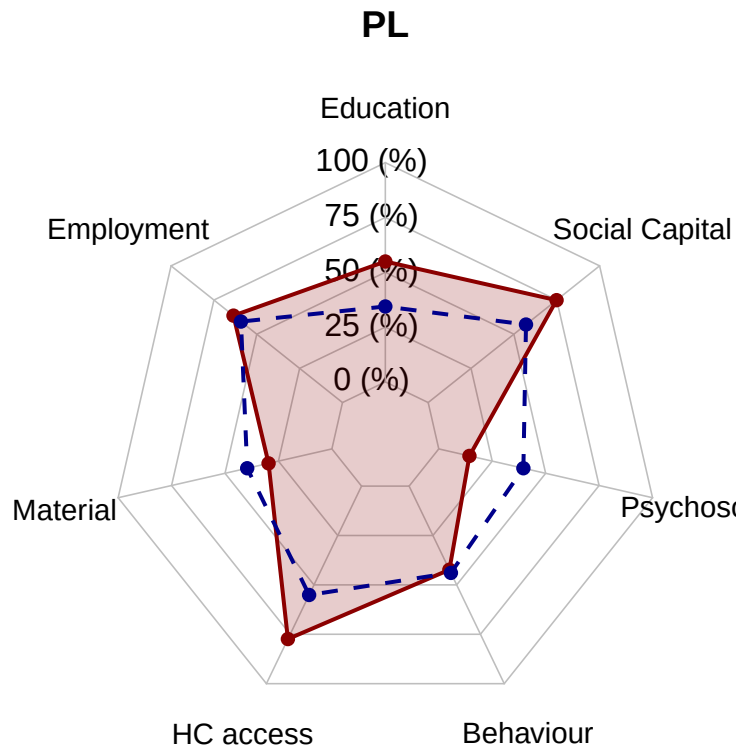


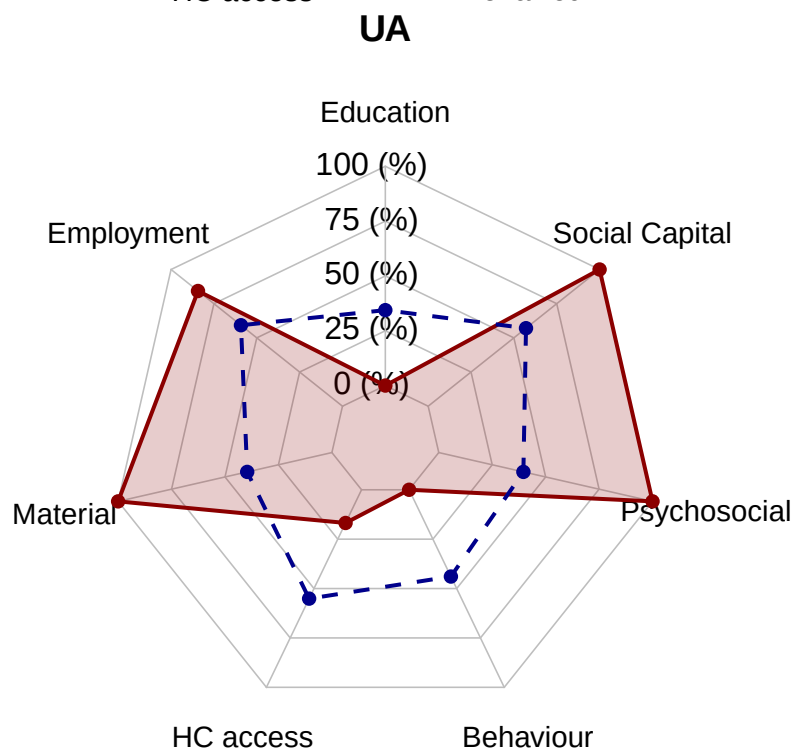
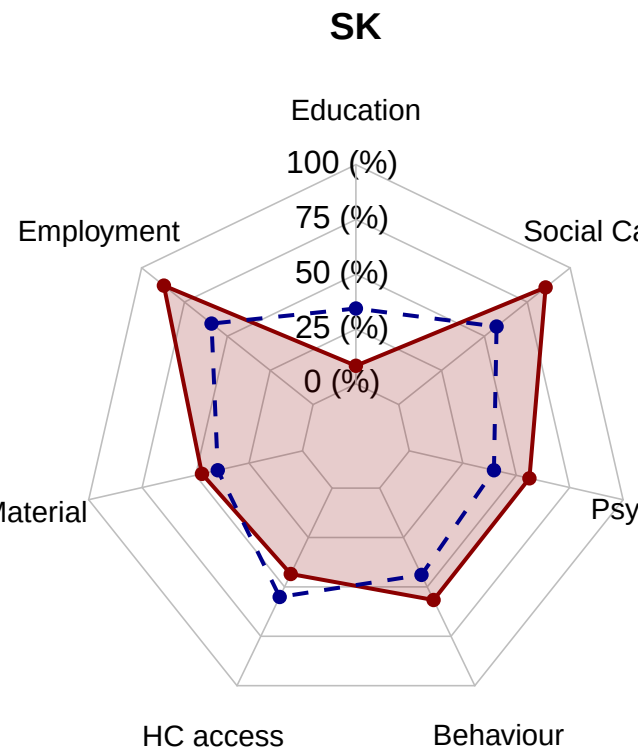
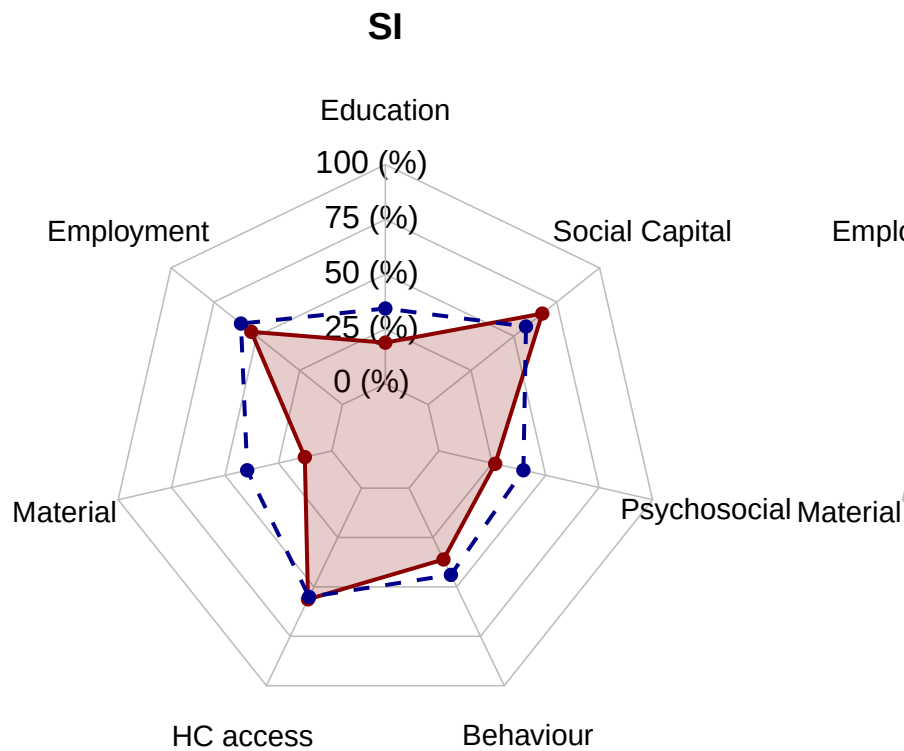












EU plots

```
# construct radar df
radarPCA_empty_df <- as.data.frame(rbind(
  pca_max_val[pca_spider_vars], ##put max/min of the whole
  pca_min_val[pca_spider_vars],
```

```

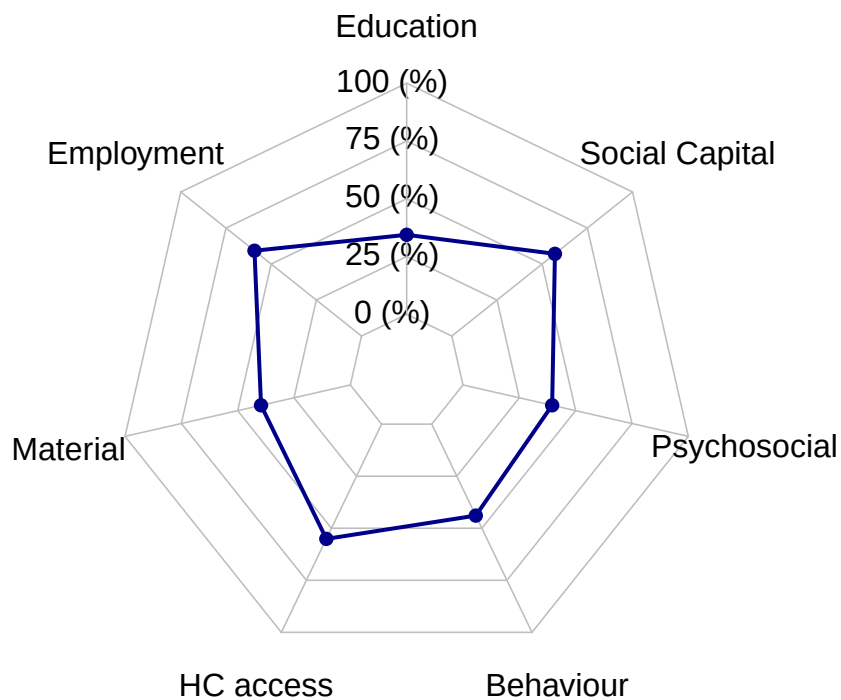
    all_pca_score_EuMean[pca_spider_vars])
)

colnames(radarPCA_empty_df) <- PCABlock_labels[pca_spider_vars]

# Show plot
par(lend = 2)
par(mar = c(1, 2, 2, 2))

radarchart(radarPCA_empty_df,
  axistype = 1,
  pcol = "darkblue",
  plwd = 2,
  cglcol = "grey",
  cglty = 1,
  axislabcol = "black",
  #caxislabels = rep("", length(caxislabels_seq)),
  cglwd = 0.8,
  vlce = 1)

```



```

# Saving plot to PNG
png(filename = paste0("Figures\\pca_per_country_spider\\Spiderplot_Empty.png"), width = 600, height = 600)
par(mar = c(1, 2, 2, 2))
radarchart(radarPCA_empty_df,
  axistype = 1,
  pcol = "darkblue",
  plwd = 4,
  cglcol = "grey",
  cglty = 1,
  axislabcol = "black",

```

```

    #caxislabels = rep("", length(caxislabels_seq)),
    cglwd = 0.8,
    vlcex = 1.8,
    cex.main = 4)
dev.off()

```

```

## pdf
## 2

```

WR plot

```

#Loop

for (WR in WR_list) {

  # Filter country-specific data
  WR_data <- allWR_pca_score %>%
    filter(regionWelfare == WR) %>%
    dplyr::select(wr_mean_edu, wr_mean_work, wr_mean_mat, wr_mean_hc, wr_mean_behav, wr_mean_psych, wr_
    as.data.frame()

  # construct radar df
  radarPCA_WR_df <- rbind(
    pca_max_val[pca_spider_vars],
    pca_min_val[pca_spider_vars],
    WR_data,
    all_pca_score_EuMean[pca_spider_vars]
  )

  colnames(radarPCA_WR_df) <- PCAblock_labels[pca_spider_vars]

  # Show plot
  par(lend = 2)
  par(mar = c(1, 2, 3, 2))

  radarchart(radarPCA_WR_df,
    axistype = 1,
    pcol = c("blue", "darkblue"),
    pfcol = c(scales::alpha("blue", 0.2), NA),
    plwd = c(2, 2),
    cglcol = "grey",
    cglty = 1,
    axislabcol = "black",
    #caxislabels = rep("", length(caxislabels_seq)),
    cglwd = 0.8,
    vlcex = 0.9,
    title = WR_labels[as.character(WR)])
  Sys.sleep(1)

  # Saving plot to PNG
  png(filename = paste0("Figures\\pca_per_country_spider\\Spiderplot_", WR, ".png"), width = 600, height = 400)
  par(mar = c(1, 2, 3, 2))

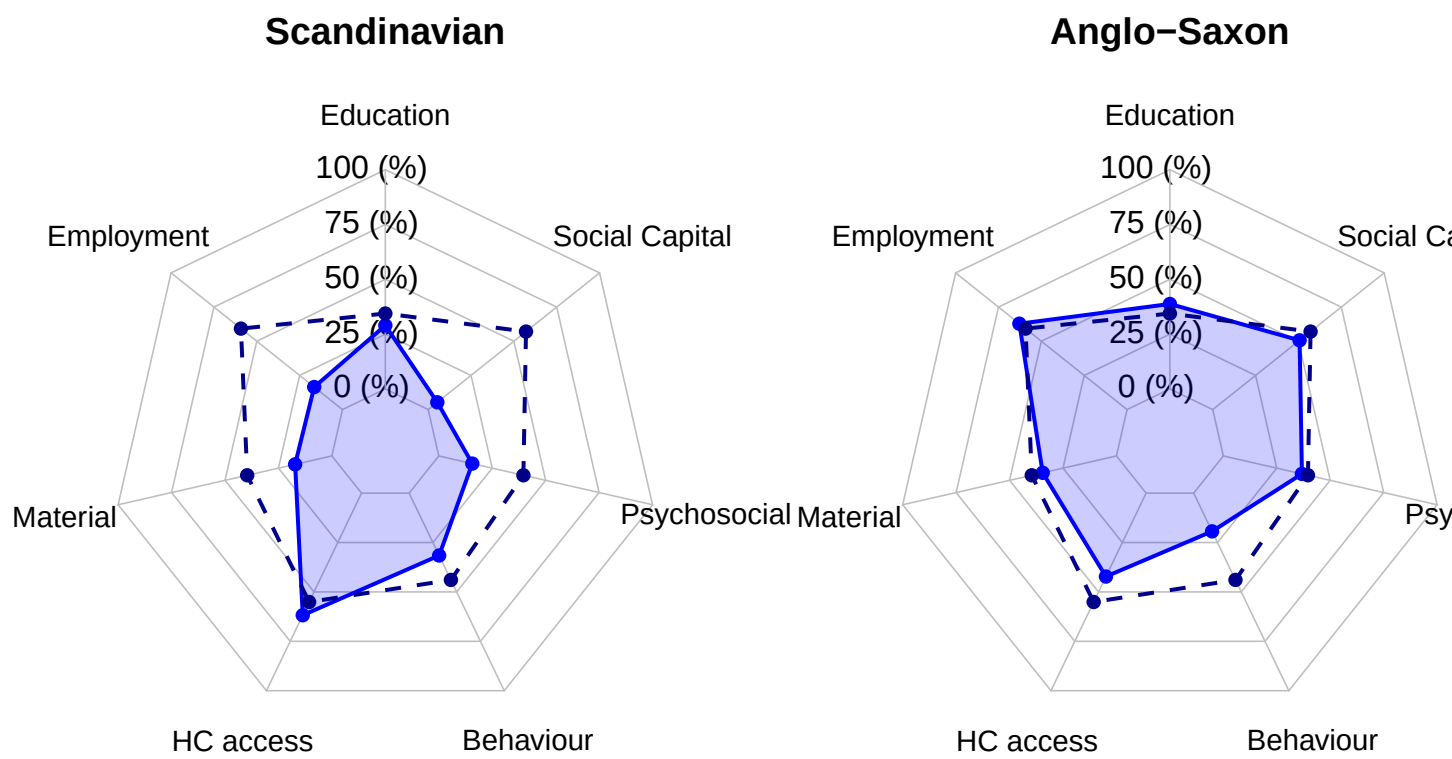
```

```

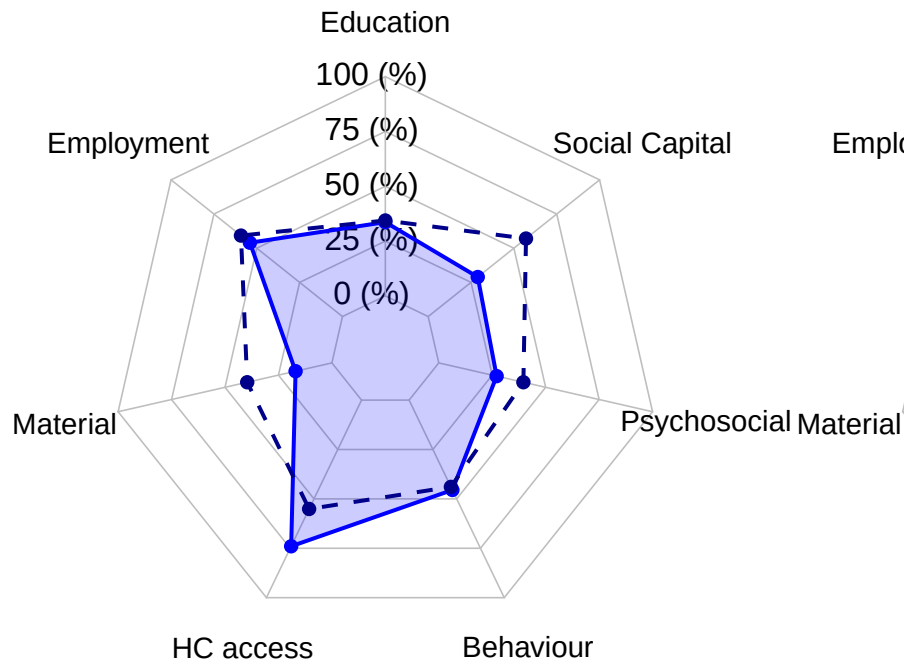
radarchart(radarPCA_WR_df,
  axistype = 1,
  pcol = c("blue", "darkblue"),
  pfcol = c(scales::alpha("blue", 0.2), NA),
  plwd = c(4, 4),
  cglcol = "grey",
  cglty = 1,
  axislabcol = "black",
  #caxislabels = rep("", length(caxislabels_seq)),
  cglwd = 0.8,
  vlce = 0.9,
  title = WR_labels[as.character(WR)],
  cex.main = 4)

dev.off()
}

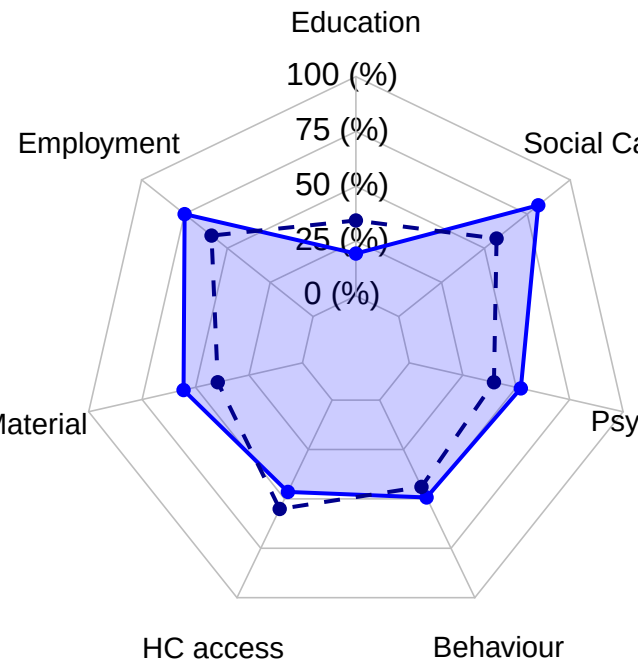
```



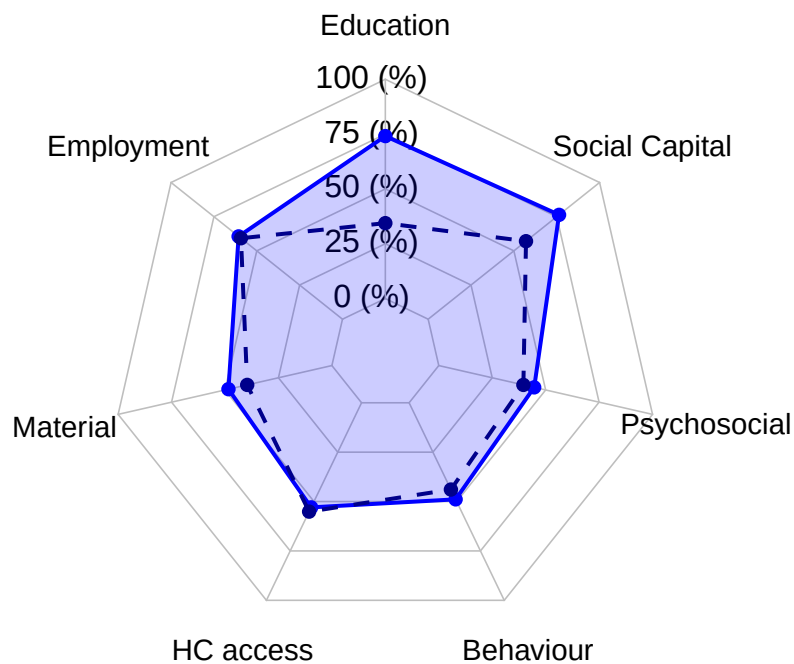
Bismarckian



Eastern European



Southern



CLUSTER ANALYSIS

```
# Dataframe to use
all_pca_score_std <- all_pca_score

# Set country names as rownames
```

```

rownames(all_pca_score_std) <- all_pca_score_std$cntry

# Keep only PCA variables and standardize
all_pca_score_std <- scale(all_pca_score_std[, 2:8])

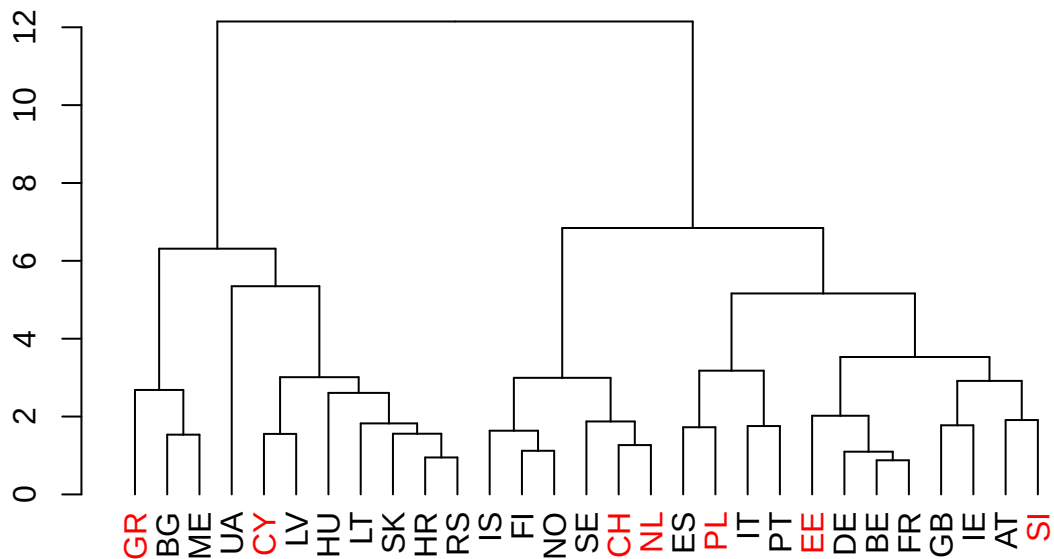
# Distance matrix
all_pca_distance <- dist(all_pca_score_std)

# Hierarchical clustering
clust_all_pca <- hclust(all_pca_distance, method = "ward.D2")

# Dendrogram
countries_wrongWR <- c("CY", "GR", "PL", "SI", "EE", "CH", "NL")

clust_all_pca_dendro <- as.dendrogram(clust_all_pca)
labels_colors(clust_all_pca_dendro) <- ifelse(labels(clust_all_pca_dendro) %in% countries_wrongWR, "red", "black")
plot(clust_all_pca_dendro,
     xlab = "", ylab = "", sub = "")

```



```

png("Figures\\Cluster\\Cluster_Ward_D2.png", width = 800, height = 600)
plot(clust_all_pca_dendro,
     xlab = "", ylab = "", sub = "")
dev.off()

```

```

## pdf
## 2

```

ANOVA

```

# Create dataset (from Spiderplot section)
all_pca_score_WR <- all_pca_score |>
  left_join(country_regionWelfare, by = "cntry")
all_pca_score_WR$regionWelfare <- as.factor(all_pca_score_WR$regionWelfare)

```

Education

```
aov_edu <- aov(index_edu ~ regionWelfare, data = all_pca_score_WR)

shapiro.test(residuals(aov_edu))

##
##  Shapiro-Wilk normality test
##
## data:  residuals(aov_edu)
## W = 0.98808, p-value = 0.98

leveneTest(index_edu ~ regionWelfare, data = all_pca_score_WR)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 4  1.6741 0.1888
##      24

summary(aov_edu)

##              Df Sum Sq Mean Sq F value    Pr(>F)
## regionWelfare  4  1.1306  0.28266    10.59 4.33e-05 ***
## Residuals      24  0.6405  0.02669
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Employment Conditions

```
aov_work <- aov(pc1_work ~ regionWelfare, data = all_pca_score_WR)

shapiro.test(residuals(aov_work))

##
##  Shapiro-Wilk normality test
##
## data:  residuals(aov_work)
## W = 0.9758, p-value = 0.7236

leveneTest(pc1_work ~ regionWelfare, data = all_pca_score_WR)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 4  1.2502 0.3167
##      24

summary(aov_work)

##              Df Sum Sq Mean Sq F value    Pr(>F)
## regionWelfare  4   5.844   1.4611    15.86 1.77e-06 ***
## Residuals      24   2.211   0.0921
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Material

```
aov_mat <- aov(pc1_mat ~ regionWelfare, data = all_pca_score_WR)
```

```
shapiro.test(residuals(aov_mat))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: residuals(aov_mat)  
## W = 0.9887, p-value = 0.9848
```

```
leveneTest(pc1_mat ~ regionWelfare, data = all_pca_score_WR)
```

```
## Levene's Test for Homogeneity of Variance (center = median)  
##      Df F value Pr(>F)  
## group 4  1.1753 0.3466  
##      24
```

```
summary(aov_mat)
```

```
##              Df Sum Sq Mean Sq F value  Pr(>F)  
## regionWelfare 4   2.476   0.6191    5.072 0.00417 **  
## Residuals     24   2.929   0.1221  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Healthcare

```
aov_hc <- aov(pc1_hc ~ regionWelfare, data = all_pca_score_WR)
```

```
shapiro.test(residuals(aov_hc))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: residuals(aov_hc)  
## W = 0.94232, p-value = 0.1153
```

```
leveneTest(pc1_hc ~ regionWelfare, data = all_pca_score_WR)
```

```
## Levene's Test for Homogeneity of Variance (center = median)  
##      Df F value Pr(>F)  
## group 4  0.3119 0.8672  
##      24
```

```
summary(aov_hc)
```

```
##              Df Sum Sq Mean Sq F value  Pr(>F)  
## regionWelfare 4  0.359 0.08976    1.321  0.291  
## Residuals     24  1.630 0.06793
```

Behavioural

```
aov_behav <- aov(pc1_behav ~ regionWelfare, data = all_pca_score_WR)
```

```
shapiro.test(residuals(aov_behav))
```



```
##
## Shapiro-Wilk normality test
##
## data: residuals(aov_behav)
## W = 0.88057, p-value = 0.003462
leveneTest(pc1_behav ~ regionWelfare, data = all_pca_score_WR)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 4  1.2894  0.302
##      24

aov_behav <- kruskal.test(pc1_behav ~ regionWelfare,
                          data = all_pca_score_WR)
summary(aov_behav)

##      Length Class  Mode
## statistic 1      -none- numeric
## parameter 1      -none- numeric
## p.value    1      -none- numeric
## method     1      -none- character
## data.name  1      -none- character
```

Psychosocial

```
aov_psych <- aov(pc1_psych ~ regionWelfare, data = all_pca_score_WR)

shapiro.test(residuals(aov_psych))

##
## Shapiro-Wilk normality test
##
## data: residuals(aov_psych)
## W = 0.98745, p-value = 0.9743
leveneTest(pc1_psych ~ regionWelfare, data = all_pca_score_WR)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 4  1.2221  0.3276
##      24

summary(aov_psych)

##      Df Sum Sq Mean Sq F value Pr(>F)
## regionWelfare 4  1.079  0.2697  2.854 0.0457 *
## Residuals    24  2.268  0.0945
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Social Capital

```
aov_sc <- aov(pc1_sc ~ regionWelfare, data = all_pca_score_WR)

shapiro.test(residuals(aov_sc))
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(aov_sc)
## W = 0.95916, p-value = 0.3134
leveneTest(pc1_sc ~ regionWelfare, data = all_pca_score_WR)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  4  1.3813 0.2701
##      24
summary(aov_sc)

##           Df Sum Sq Mean Sq F value   Pr(>F)
## regionWelfare  4 10.243   2.5609   33.42 1.73e-09 ***
## Residuals    24   1.839   0.0766
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```